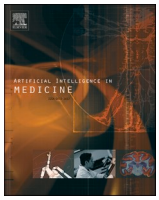
Artificial Intelligence In Medicine 127 (2022) 102285



Contents lists available at ScienceDirect

Artificial Intelligence In Medicine

journal homepage: www.elsevier.com/locate/artmed



Research paper



BreastScreening-AI: Evaluating medical intelligent agents for human-AI interactions

Francisco Maria Calisto [a,*], Carlos Santiago [a], Nuno Nunes [b], Jacinto C. Nascimento [a]

[a] Institute for Systems and Robotics, Avenida Rovisco Pais 1, 1049-001 Lisbon, Portugal
[b] Interactive Technologies Institute, Caminho da Penteada, 9020-105 Funchal, Madeira, Portugal

ARTICLE INFO

Keywords:
Human-computer interaction
Artificial intelligence
Healthcare
Medical imaging
Breast Cancer

ABSTRACT

In this paper, we developed *BreastScreening-AI* within two scenarios for the classification of multimodal beast images: (1) *Clinician-Only*; and (2) *Clinician-AI*. The novelty relies on the introduction of a deep learning method into a real clinical workflow for medical imaging diagnosis. We attempt to address three high-level goals in the two above scenarios. Concretely, how clinicians: i) accept and interact with these systems, revealing whether are explanations and functionalities required; ii) are receptive to the introduction of AI-assisted systems, by providing benefits from mitigating the clinical error; and iii) are affected by the AI assistance. We conduct an extensive evaluation embracing the following experimental stages: (a) patient selection with different severities, (b) qualitative and quantitative analysis for the chosen patients under the two different scenarios. We address the high-level goals through a real-world case study of 45 clinicians from nine institutions. We compare the diagnostic and observe the superiority of the *Clinician-AI* scenario, as we obtained a decrease of 27% for False-Positives and 4% for False-Negatives. Through an extensive experimental study, we conclude that the proposed design techniques positively impact the expectations and perceptive satisfaction of 91% clinicians, while decreasing the time-to-diagnose by 3 min per patient.

1. Background

Recently, Deep Learning (DL) methods started outperforming traditional machine learning algorithms in many application domains, when the access to large amount of data and training *datasets* was feasible [37,126]. In this work, we introduce a DL model into a multi-modal breast imaging system. We developed a medical imaging framework called *BreastScreening-AI* that allows radiologists to visualize and manipulate images, while also accessing a diagnostic recommendation from an AI assistant. By using the framework, clinicians can accept or reject the assistant proposed severity classification. The framework is used to test the usefulness of an intelligent system that is integrated into a real-world clinical scenario.

Most physicians do not expect their clinical systems to behave inconsistently and imperfectly [69,86], which can lead to mistrust and potential abandonment of these technologies within real-world clinical setups [19,25,26]. While considerable work as focused on improving the accuracy of AI algorithms, comparatively less work focused on improving trust and usability of interactive assistance techniques. Therefore, we ask several high-level questions, namely: i) how they *accept* and *interact* with these systems; ii) how *receptive* are the clinicians to the introduction of AI assistance; iii) how clinicians are *affected* by the AI assistance in different clinical contexts. This paper contributes broadly to the literature in Human-Computer Interaction (HCI) and Artificial Intelligence (AI) by examining what radiologists need when using AI-powered image diagnostic, the practices they adopt while using these tools, and how they affect end-user attitudes towards the underlying AI algorithms [3,107].

The fields of HCI and AI are intersecting within several DL work models to assist clinical practitioners with appropriate knowledge [97,101,123]. However, the literature does not typically address the effectiveness of clinical AI solutions from the perspective of human users [29,78]. The work of Xie et al. [139] has established the importance of providing reasons for AI recommendations to foster model transparency and user trust. This work stressed that clinical systems should be designed based on the user needs of clinical practitioners.

Schaekermann et al. [125] argue that explanations for clinical cases can be leveraged by AI assistants to support medical reasoning. Although similar to what we are doing in this work, these authors are not using real AI outputs. Instead, they are simulating hypothetical AI

* Corresponding author.
E-mail address: francisco.calisto@tecnico.ulisboa.pt (F.M. Calisto).

https://doi.org/10.1016/j.artmed.2022.102285
Received 15 February 2021; Received in revised form 15 March 2022; Accepted 21 March 2022
Available online 29 March 2022
0933-3657/© 2022 The Authors. Published by Elsevier B.V. This is an open access article under the CC BY-NC-ND license (http://creativecommons.org/licenses/by-nc-nd/4.0/).

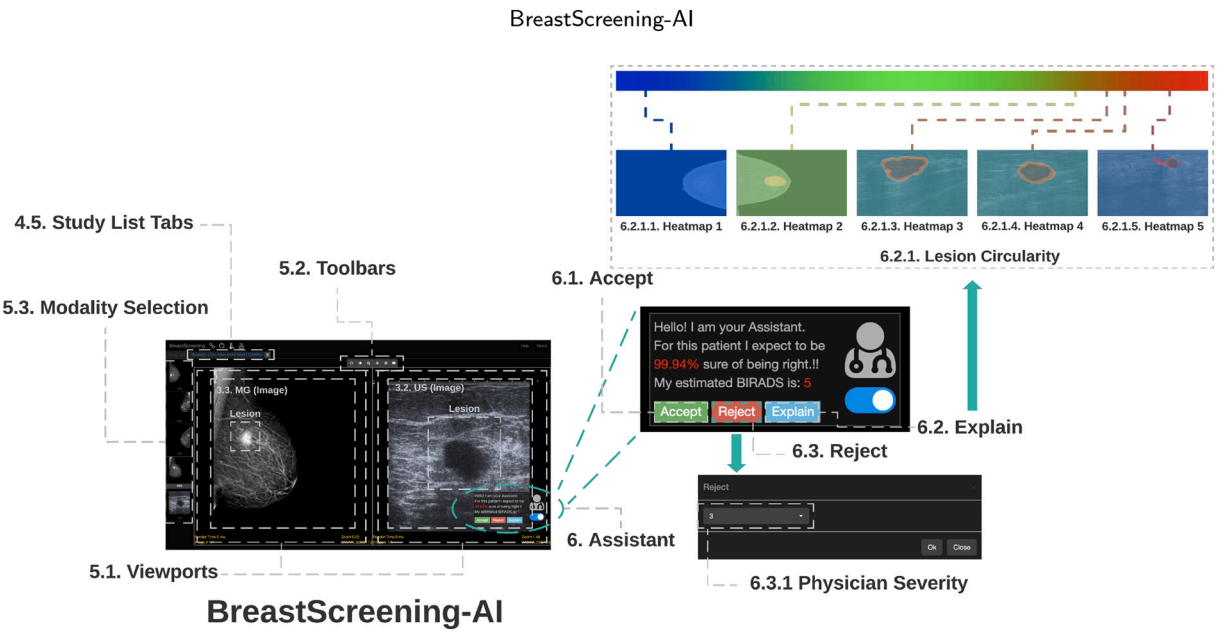


Fig. 1. Our *BreastScreening-AI* provides several functionalities. Specifically, it is possible to validate the DenseNet output classification along with radiologists. The system is framed in the following subcategories: 4.5. Study List Tabs; 5.1. Viewports; 5.2. Toolbars; 5.3. Modality Selection; 6. Assistant; 6.1. Accept; 6.2. Explain; 6.3. Reject; and 6.3.1. Physician Severity. On 6.2. Explain, the assistant will pop-up several heatmaps.

scenarios and agent behaviors, manually generated by radiologists. Our work contributes with novel perspectives on the system design and integration of a real AI model for clinical assessments.

These bodies of work demonstrated that model interpretations should be formulated in the context of clinical knowledge of a real setting. Indeed, as users require medical explanations during adoption [56,93,140], the scientific community should adapt medical explanations depending on specific clinical domains and real clinical scenarios [78,92,106]. Similarly, our work extensively investigates the desirable characteristics of AI-based assistance for breast cancer diagnosis and proposes how these functionalities can be more reliable for a proper decision-making.

Previous works, outside the clinical scope [53,86], denote that the impact of False-Positive (FP) *vs.* False-Negative (FN) on UX is generally unexplored. However, this is of high relevance when considering AI for the clinical domain [22,50] as it will be experimentally shown. Indeed, measuring predictions are typically quantified as precision in contrast with recall [128]. We, therefore, explore the following Research Questions and associated each to the set of Hypothesis following the guidelines described in the literature [8,86].

Specifically, we consider the following three research questions and related hypothesis:

- **RQ1.** What is the impact of an AI system on the medical workflow?
  An AI system focused on Precision/Recall will result:
  - **H1.1.** in higher accuracy.
  - **H1.2.** in higher time performance.
- **RQ2.** Will explainability increase clinicians' expectation and perception over an AI recommendation?
  Providing an explanation will improve clinician:
  - **H2.1.** understanding.
  - **H2.2.** trust.
  - **H2.3.** acceptance.
- **RQ3.** How does an AI system affect the inter-variability and intra-variability of clinicians during patient classification?
  - **H3.1.** The inter-variability is improved by the introduction of AI.
  - **H3.2.** The intra-**variability** is improved by the introduction of AI.

RQ1. explores the impact of our assistant in avoiding medical errors concerning clinicians' accuracy and performance. **RQ2.** evaluates the expectation in increasing perceptive satisfaction with the system, while providing an explanation, to improve understanding, trust, and acceptance of clinicians over the AI recommendation. Finally, **RQ3.** addresses the clinicians' inter-variability and intra-variability during patient classification, without and with AI recommendations. The above RQs are placed in a context of Clinician-Only *vs.* Clinician-AI scenarios discussed in the evaluation section.

1.1. Clinical workflow

In breast cancer, medical imaging systems allow the end-user to diagnose several modalities, such as MammoGraphy (MG), UltraSound (US) or Magnetic Resonance Imaging (MRI), from the retrieval of medical imaging data [58,127]. A wide range of Clinical Decision Support Systems (CDSS) exist, providing clinicians with the knowledge to enhance the clinical workflow [28,55,64], from systems that supply potential information for medical decision-making to those that make diagnostic decisions [77,111].

Experts may resist using a system if it does not capture the nuances of their mental models or provides relevant context [84,87,141]. Conversely, clinicians are known to resist changes and new tools [35,60], an issue that has an impact in their workflow. We focus on understanding different aspects and expectations of a medical imaging CDSS integrated into the radiology workflow. In particular, our work demonstrates how an interactive AI can directly address the above-mentioned issues during medical imaging diagnosis.

1.2. Commercial solutions

Image algorithms and computer aids to assist the clinician have been applied to a number of clinical tasks, though the focus has been on automated quantitative of biomarkers [83]. Some researchers have developed a commercial solution, called PAGODA [110]. Yet, this software does not provide a radiologic solution, nor it is ready for the pipeline of breast cancer diagnosis [37].

Other interesting solutions are the NIRAMAI [21] and CureMetrix

[98]. In these solutions, respectively thermal and mammography images are used for diagnostic. However, the multimodality is not used as we proposed in this paper.

1.3. Explainability & interpretability

Concerning explainability and interpretability, two approaches are currently proposed to address this challenge: eXplainable-AI (XAI) and Intelligibility [65,103]. XAI and Intelligibility must take into account the fact that diverse data may contribute to a relevant result [13,20]. The work done by Holzinger et al. [72] addresses this need, namely *how* and *why* a machine has reached a given decision [129]. Moreover, transparent algorithms could appropriately enhance the trust of clinicians in future Human-AI interactions [52,137].

Whilst XAI [122] deals with the implementation of transparency and traceability of statistical 'black-box' for the ML methods [2]. Particularly, DL approaches are in certain domains, a pressing need to go beyond XAI; for example, to reach a level of explainable medicine there is a crucial need for causability [74]. However, causability [73] is different from causality.

In the same way that usability encompasses measurements for the quality of use [4,133], causability encompasses measurements for the quality of explanations [71] produced by XAI methods (*e.g.*, the heatmaps from the *Lesion Circularity* of Fig. 1). Specifically, causability is the property of a human (*i.e.*, human intelligence), whereas explainability is a property of a system (*i.e.*, AI).

1.4. Human-AI interaction

Although the research in interaction with intelligent systems is recent [24,82], still this topic has seen new advances *e.g.*, chat-bots and other agents [43,102,104]. Recent advances in medical technologies that promote the generation of data, have continued to drive interaction research in the clinical domain [15,46]. Moreover, the new interest of the medical community to support AI research projects and the available public *datasets*, are encouraging researchers to work on both fields [88,114]. In this paper, we build on a growing interest in studying the broader, global aspects of model transparency, and conduct a deep exploration of these issues within the domain of decision-making in medical imaging.

Human-AI Interaction (HAII) incorporates human feedback in the model training process to create better ML models [18,95]. The topic is also known as Interactive Machine Learning (iML) [57] or Human-In-The-Loop (HITL) [70]. In this paper, we refer to the topic as HAII, which is addressed in Amershi et al. [8] providing a set of design guidelines.

In Kocielnik et al. [86], it is also addressed a study for the impact of several methods of expectation setting, and others studied the design for specific HAII scenarios [5]. While much of the mentioned prior work has employed handcrafted features [8,86], we leverage the rich image data features automatically learned from DL algorithms. By focusing on the clinical domain, we argue that HAII research could improve clinical recommendations and clinical decision-making [75].

1.5. Decision-making process

Giving clinicians access to AI systems, requires mechanisms to understand the machine decision process [79,135]. While we can assume that expert clinicians are always able to explain their decisions, it is interesting to observe that sometimes they are not able to provide a clear explanation. One example is the vast amount of data sources and their diversity [20,88,114]. Consequently, HAII calls for confidence and trust, bringing usability into a new and vital focus of AI.

Expectation theories are postulating that user satisfaction and acceptance of a system is directly related to the difference between initial expectations and the actual clinical experience [86,91].

Table 1
The 18 human-AI interaction design guidelines [8]. These guidelines were validated through multiple rounds of evaluation. We mapped the guidelines to each research question. Specifically, the **RQ1** was mapped with **G3**, **G7**, **G8**, **G9**, **G12**, **G13**, **G14**, **G15**, **G16**, and **G18** guidelines. The **RQ2** was mapped with **G1**, **G2**, **G4**, **G10**, **G11**, and **G17** guidelines. Finally, the **RQ3** was mapped with **G5**, and **G6** guidelines.

AI Design Guidelines			
G1	Make clear what the system can do.	G10	Scope services when in doubt.
G2	Make clear how well the system can do what it can do.	G11	Make clear why the system did what it did.
G3	Time services based on context.	G12	Remember recent interactions.
G4	Show contextually relevant information.	G13	Learn from user behavior.
G5	Match relevant social norms.	G14	Update and adapt cautiously.
G6	Mitigate social biases.	G15	Encourage granular feedback.
G7	Support efficient invocation.	G16	Convey the consequences of user actions.
G8	Support efficient dismissal.	G17	Provide global controls.
G9	Support efficient correction.	G18	Notify users about changes.

Specifically, expecting more than the system can deliver will decrease user satisfaction and lead to the rejection of the system [115]. Hence, we created our proposed design techniques, based on the following contributions: (1) providing users a new control feature on the introduction of AI methods among medical imaging diagnosis; and (2) the impact of the users' *behavior* and the impact in professional practice. We did that to achieve more accurate expectations of the systems capabilities, addressing potential gaps.

2. BreastScreening framework

The proposed *BreastScreening* framework incorporates an assisted tool, as it is able to integrate an AI model for breast cancer classification and segmentation. This assisted tool offers radiologists an autonomous second reader opinion during the breast cancer diagnosis. Specifically, we are using a DenseNet model [45].

To validate the proposed DenseNet results, our framework allows the radiologist to *accept* or *reject* the suggested BI-RADS classification [61]. Therefore, the radiologist can freely control the final diagnosis result [38,144]. The framework operates as a website accessed *via* web browser.

2.1. Design choices

The work developed by Amershi et al. [8], describes a set of 18 guidelines (Table 1) for Human-AI Interaction (HAII) being highly useful to support our research questions. Also, in [86] it is provided an exploratory study of an assistant to study the impact of several methods for user expectations. In both studies, the authors show that different focus on avoiding types of errors lead to a different subjective perception of accuracy and acceptance.

Prior work [86] shows three significant contributions to clinician's expectations: (1) information from external sources; (2) reasoning and understanding; and (3) first-hand experience. From here, we aim to explore several design techniques for achieving clinicians' expectations by following the set of guidelines [8].

Due to the multi-view and multi-modal nature of the data, our assistant uses large amounts of images [39,136]. The most common used modalities are as follows: (i) MG (both CC and MLO views); (ii) US; and (iii) MRI volumes. The above modalities results from a selection provided by the radiologists team. Notice that in (i) a large number of MG views are available, *e.g.*, ML, LM, LMO, late ML, among others. Concerning (iii), the acquisition allows obtaining a large set of MRI data (*e.g.*, T1, T2, Diffusion, DCE-MRI) [127].

Initial design choices resulted in a consensus that allowed the

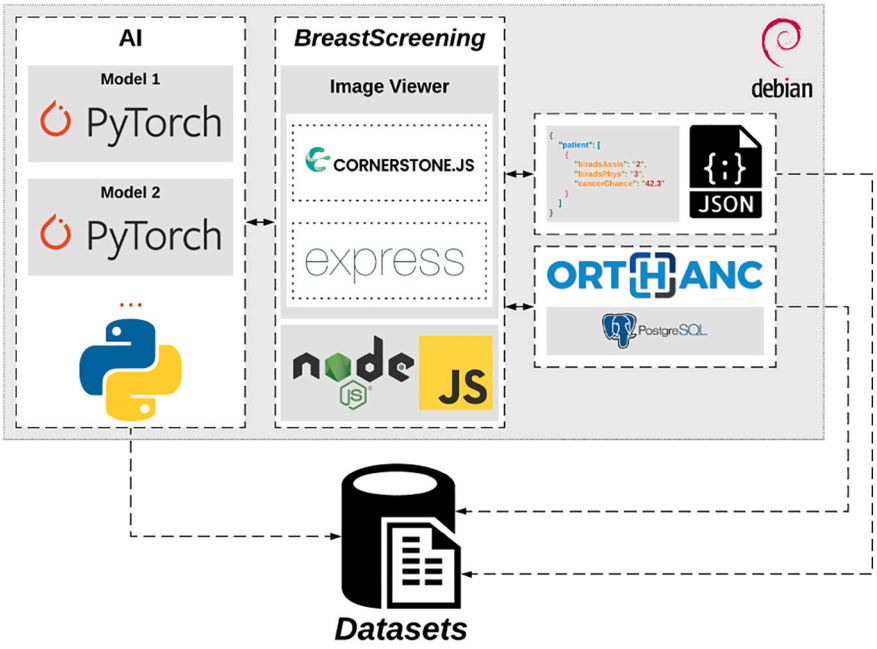


Fig. 2. *BreastScreening-AI* Architecture: the main components of the system are AI Application, Image Viewer, DICOM Storage. The Image Viewer will provide essential interaction tools for radiologists.

selection of CC and MLO views, in the case of MG, and DCE-MRI. From the above, it is vital to have the *5.1. Viewports* management functionality (Fig. 1), providing the users with the opportunity to arrange the viewport depending on the available modalities and views. Furthermore, we choose to implement the *5.2. Toolbars* with a quick and configurable *5.1. Viewports* management button, to support this use case specifically for breast cancer.

Our evaluation (Section 4) validated design choices and revealed clinician strategies for using the assistant. The learned design lessons should, theoretically, generalize to other healthcare assisted systems [11,89]. Building upon these design lessons, we focus on exploring human-AI collaboration in the context of clinical decision-making. Next, we present our design choices based on the observations and information from the participatory sessions with clinicians.

2.2. User interface

Based on the user needs, we designed the *BreastScreening-AI* framework. The *BreastScreening-AI* framework is including a set of refinement mechanisms to guide radiologists during the diagnostic process. We integrated the assistant in the User Interface (UI) of an early version (with no AI) of the framework [36], namely *BreastScreening*, consisting of three main functionalities, further described (Section 7.9).

The *4.5. Study List Tabs* (Fig. 1) gives radiologists the opportunity to switch between the patient who is being diagnosed and a full list of patients. Using the *5.2. Toolbars* on the *5.1. Viewports*, the clinician can locate the lesions and classify its severity (*via* BI-RADS), choosing the *6.1. Accept* or *6.3. Reject* options for the given severity of the *6. Assistant*.

When the several modalities [90] are correctly used (regarding the *5.3. Modality Selection* on a multimodality view), the clinician can find more accurately the right severity classification. For the *6.3. Reject* option, the clinician will have to insert the proposed BI-RADS on a drag-and-drop menu (*6.3.1. Physician Severity*) of severity options. Finally, the clinician may look for the *6.2. Explain* functionality (Fig. 1), which provides an explanation regarding the *Lesion Circularity* values [109].

The heatmaps are obtained from the manual annotations [31,36]. From these annotations, we compute the *Lesion Circularity* and color-related (Fig. 1) to the computed severity. The generation of the above color maps come from the following information: (i) the area (Lesion Size) of the lesion that comes from the delineation process, (ii) the circularity/sharpness (Lesion Shape) that can be computed from the annotation in (i), and (iii) the BI-RADS score automatically provided by the DenseNet (*i.e.*, classifier assistant).

2.3. Implementation

BreastScreening framework was implemented using *CornerstoneJS* (cornerstonejs.org) [134] with *NodeJS* (nodejs.org) [54,59] and *ExpressJS* (expressjs.com) [66] for managing the server-side. To feed the system, we selected image sets from HFF and uploaded them into an *Orthanc* server [81]. We collected the patient data from this first institution (*i.e.*, HFF). Then, we applied our User Testing and Analysis (UTA) [32] to the remaining eight clinical institutions.

Three imaging modalities (MG, US, and MRI) were required from each patient. We pre-processed the images and anonymized on the *Orthanc* server, then retrieved by the *BreastScreening-AI* framework. We efficiently designed this system as a set of modules (Fig. 2) that can be reused in other imaging applications.

The *CornerstoneJS* family of libraries provide essential functions, such as i) image rendering; ii) DICOM retrieval; iii) tool support (*i.e.*, developed functionalities); and iv) interpretation. Additionally, the *BreastScreening* core was developed in *JavaScript* with *jQuery* (jquery.com) [49] for *HTML* manipulation, event handling of the US interactions, such as mouse events, and look & feel respectively.

Additionally, we used the *dicomParser* [142] for parsing (github.com/cornerstonejs/dicomParser) DICOM files. The DICOM files can be loaded by drag-and-drop into the browser window on the *Orthanc* view. Loaded images can be further displayed on both views, but with different visualization configurations. After loading stage, images are automatically arranged according to the scan IDs from the DICOM files.

Finally, for the assistant, the DenseNet was developed using PyTorch [113]. PyTorch is a deep learning library that is widely used by the machine learning community. Due to its imperative and "Pythonic" programming style, the library showed to be reliable to this work.

2.4. Medical imaging datasets

In order to control the medical imaging information, radiologists

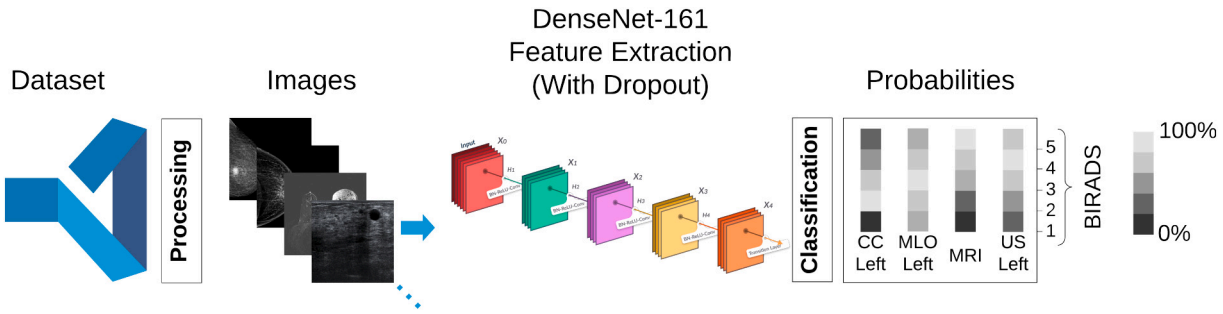


Fig. 3. DenseNet extracts features from images using dense blocks to efficiently compute the probability of each class.

interact with in a sub-sequence study. The *BreastScreening* was fixed to operate on a limited subset of 289 classified patients from the collected *dataset* at Hospital Prof. Doutor Fernando Fonseca (HFF). In fact, from our full *dataset* of medical images contains more than 200,000 DICOM files from a total of 338 patients.

Each of the 289 patients was classified by the head of radiology of Medical Imaging Department at HFF. The expert provide us both BI-RADS values of the pre-biopsy and pathological results of each patient.

From the pre-biopsy classification, we divided the *dataset* into three distinct patient types: **P1** with low severity, *i.e.*, BI-RADS ≤ 1; **P2** with medium severity, *i.e.*, $1 <$ BI-RADS ≤ 3; and **P3** with high severity, *i.e.*, BI-RADS > 3; on all possible modalities (MG, US and MRI). Each radiologist will open each patient (*e.g.*, **P1**, **P2** or **P3**), that is chosen randomly, and will examine the set of images.

The *dataset* resulted from work done by eight of the 45 clinicians. All of the eight clinicians are resident radiologists of HFF. These clinicians are working directly with the head of radiology of the HFF institution.

2.5. Model description

A fundamental component in our UI is the integration of a deep neural network (*i.e.*, DenseNet). A DenseNet is capable to automatically provide the classification for the lesion severity of the exam. For our study, the DenseNet [76] architecture (Section 7) used in *BreastScreening-AI* was DenseNet-161 (Fig. 3). The DenseNet was initially pretrained on ImageNet [48], a large dataset of 1.2 million images from 1000 classes. This is an important step to regularize the network whenever small sets are available for training. Then, we fine-tune the network parameters using our medical dataset. This is accomplished by removing the last layer and replacing it by a new fully-connected layer with five classes, corresponding to the BI-RADS values (from 0 to 5).

For fairness purposes, we split our dataset into two disjoint sets: training and testing sets. The training is used to fine-tune the weights of the architecture to our medical dataset. The test is a hold-out set used exclusively for testing. Notice that, in the experimental evaluation, only the test set is used, where the radiologists are asked to perform the diagnosis. All modalities were evaluated by the same model, and required a pre-processing step, which included resizing the images to 224×224 pixels (with bilinear interpolation), as well as normalization which removed the mean and standard deviation, similarly to what was done for ImageNet. Training was performed using the Adam optimizer [85], with the default parameters (learning rate of 10^{-3} and weight decay of 10^{-4}).

Online data augmentation was also used to prevent over-fitting, namely by using random cropping (with a 10 pixel zero-padding on each side) and random horizontal flipping. To train the model, the cross-entropy loss was used, in which the model predictions (after softmax) were compared to the classifications provided by the team of radiologists. At test time, the unseen images went through the same pre-processing step mentioned above. Both the model predictions (*i.e.*, the probability assigned to each class, which expresses the model confidence in that classification) and the classification (most probable class) were stored. With this setup, the model achieved an accuracy of 94.81% on the multimodal test set.

3. Methods

We evaluated *BreastScreening-AI* simulating real-world conditions with 45 clinicians in nine different clinical institutions. Our goal was to quantitatively and qualitatively assess the proposed design principles that the *BreastScreening-AI* system embodies (rather than on particular widgets of the UI) and to understand how these principles would fare in practice [12].

We were particularly interested in understanding how *BreastScreening-AI* would accomplish improvements of diagnosis, surpassing the challenges [112] and providing evidence of the opportunities emerging from HAII [29,125]. Ultimately, we were focused on radiologists' accuracy while using our *BreastScreening-AI* functionalities and on how to improve AI reliability during medical decision-making [4,7].

We have collected a large breast imaging database. The database contains 338 multi-modal image cases. Specifically, it contains mammography, ultrasound and magnetic resonance imaging. Also, our database contains the manual delineation of the lesion (*i.e.*, ground-truth) along with the classification of the lesion severity that is indexed on the BI-RADS [61]. This is required as the BI-RADS reporting enables radiologists to communicate results.

3.1. Participants

We asked participants to practice with three predefined patients (*i.e.*, **P1**, **P2** and **P3**). Then, we asked participants to diagnose each patient. Recall that these patients belong to the test set, that have never been seen by the DenseNet. We expected the assistant to minimize the time required and accuracy (*e.g.*, reducing the values of False-Positives or False-Negatives).

Our study involved 45 clinicians, recruited on a volunteer basis from a broad range of clinical scenarios, including nine different health institutions (two public hospitals, two cancer institutes and two private clinics). From the demographic questionnaires: 24.44% of the clinicians have between 31 and 40 years of practical experience (Seniors); 31.11% have between 11 and 30 years of experience (Middles); 17.78% have between 6 and 10 years of experience (Juniors); and 26.67% have limited experience (Interns).

Interviews were conducted in a semi-structured fashion, taking about 30 min per clinician. Our goal with these semi-structured interviews was to garner their clinicians' the AI use, barriers and enablers to their use in clinical practice. The interviews were transcribed verbatim and thematically analyzed. Overall, the team spent 57 days on the clinical

institutions[1] for the observation process and six months for the classification.

3.2. Apparatus

To track user interactions across our system, we used Hotjar [94]. This tool is an analytic package allowing us to follow our users remotely. It also provides two critical pieces of functionality, among others, that can aid in remote user testing. First, the frequency areas allow us to see where users are clicking, tapping and scrolling on our system. Second, it records a video playback of the entire user session. The tool shows evidence of being useful for the studies conducted herein. To record the task activities and the interview, we used QuickTime [120] for user's screen recording.

Finally, to understand radiologists' reading behavior we took advantage of an eye-tracking device. The selected device was the Tobii Eye Tracker 4C, since Tobii is the largest commercial eye tracking manufacturer, and is one of the most common trackers among usability practitioners [130]. Also, we used the Tobii Pro SDK [42], providing us with the gaze information of the eye tracking device.

3.3. Analysis

For the study, we conducted both quantitative and qualitative analysis. The quantitative analysis focused on measuring the radiologists' performance (*e.g.*, number of False-Positives and False-Negatives) during diagnosis of three groups of patients (*i.e.*, low, medium and high severity) for each doctor. The above measurements are part of the quantitative analysis with a comparison between *Clinician-Only* and *Clinician-AI* setups.

To do a detailed comparison, we will answer the research questions and hypotheses mentioned in Section 1, providing evidence for the impact, increased expectations and perception of our system, as well as how does AI assistance is affecting the radiology workflow. We used the *BreastScreening-AI* prototype as a medium to engage clinicians and observe how they perform diagnosis with AI involved in the process. Additionally, we aim to elicit qualitative reactions of clinicians, as well as feedback. For the qualitative analysis, we extract opinion-based feedback from the recorded audio during the interaction with the prototype. The content analysis that led to a set of key-phrases expressing the number of clinicians who shared similar opinions.

3.4. Procedure

In our study, we used a within-subjects design to compare the performance of both *Clinician-Only* and *Clinician-AI* setups. Succinctly, clinicians first assess the three patients (*i.e.*, spread by Low, Medium and High severities) on the *Clinician-Only* setup and then, with the other three patients, on the *Clinician-AI* setup. We derived tasks from test scenarios developed from use cases and/or with the assistance of a subject-matter expert. The tasks are identical for all participants of a given user role in the study.

Each participant interacts with the assistant, *accepting* or *rejecting* the system suggestion, that is, the output classification of the deep neural network. The test set will comprise a number of 289 patients, where each patient in the set must have at least one of the three available modalities. Each participant will open the respective set of three patients (*e.g.*, **P1**, **P2** or **P3**), chosen randomly, and will examine the set of images. During the examination, the participant will interact with the available features of the system.

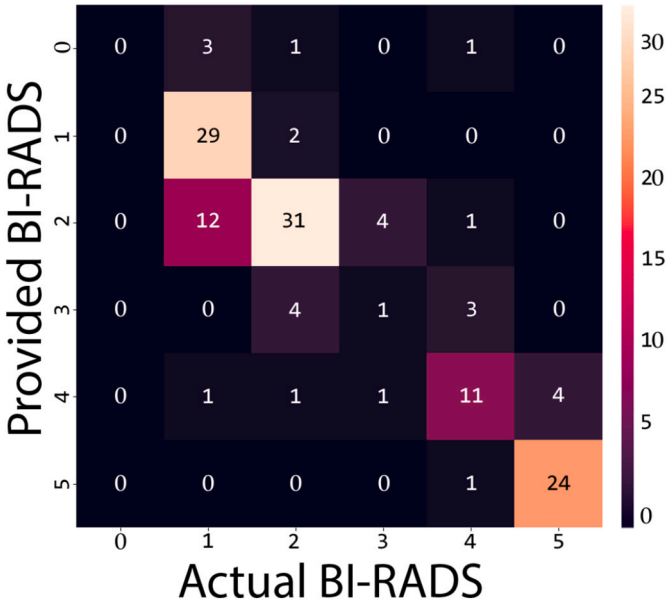


Fig. 4. Confusion Matrix with our information of the collaboration (*Provided*) between clinicians and our AI (*Clinician-AI*) technique. For this case, we used our DenseNet results to compute the BI-RADS outputs supporting the physician decision. The classification accuracy of clinicians with AI recommendations was 71% for the number of True-Positives. On the other hand, the number of False-Negatives was 2% and the number of False-Positives was 27%, only. The *Provided* value was most accurate in classifying low (BI-RADS < 2) and high (BI-RADS ≥ 4) severity cases. The columns represent the *Actual* (biopsy confirmed) category of the objects and the rows represent the *Provided* (collaboration between physician and AI) value.

3.5. Quantitative measures

We measured the perceived confidence of clinicians while interacting with the system through three questions adapted from the model of trust [27] that we called Dimensions Of Trust Scale (DOTS): (1) **Understanding**: "*I understand what the system is thinking*"; (2) **Capability**: "*The system seems capable*"; and (3) **Benevolence**: "*The system seems benevolent*". The three questions were answered on a 20-point scale from "*0% - Totally Disagree*" to "*100% - Totally Agree*" with 5% increments. We also did *post-task* questions.

The *post-task* questions, aim to evaluate satisfaction, were adapted from [86]: *How well the assistant worked?*, that is important to the adapted model of trust [27]. We also measure the recall and precision of the system. We then applied a standard analysis to determine the effect of the attributes on the clinician's interpretation and expectations.

In order to have a fair measure of the system performance [100], we use the Ground-Truth (GT) the real BI-RADS values (provided by the head of the radiology department). From the GT, we compute the number of True-Positives (TP), True-Negatives (TN), False-Positives (FP) and False-Negative (FN). We measured accuracy as the percentage of correctly classified patients.

In order to take further conclusions of our data, we summarize the following performance measures: (a) Recall; and (b) Precision. For instance, if a clinician provides a BI-RADS of 3 but the real BI-RADS is a 5 we consider it as an FN result. On the other hand, if the real BI-RADS is a 2, but the clinician provides a BI-RADS of 4 we consider it as an FP

[1] Number of clinicians and institutions that supported our project: 12 clinicians of Hospital Professor Doutor Fernando Fonseca (HFF); 10 clinicians of Instituto Português de Oncologia (IPO) de Lisboa (IPO-Lisboa); 2 clinicians of Hospital de Santa Maria (HSM); 9 clinicians of IPO de Coimbra (IPO-Coimbra); 1 clinician of Madeira Medical Center (MMC); and 1 clinician of Serviços de Assistência Médico-Social do Sindicato dos Bancários do Sul e Ilhas (SAMS); 8 clinicians of Hospital do Barreiro (HB); 1 clinician of Hospital de Santo António (HSA); 1 clinician of João Carlos Costa Diagnostic Imaging (JCC). We collected the patient data from the first institution (*i.e.*, HFF). Then, we applied our User Testing and Analysis (UTA) to the full list of clinical institutions and clinicians.

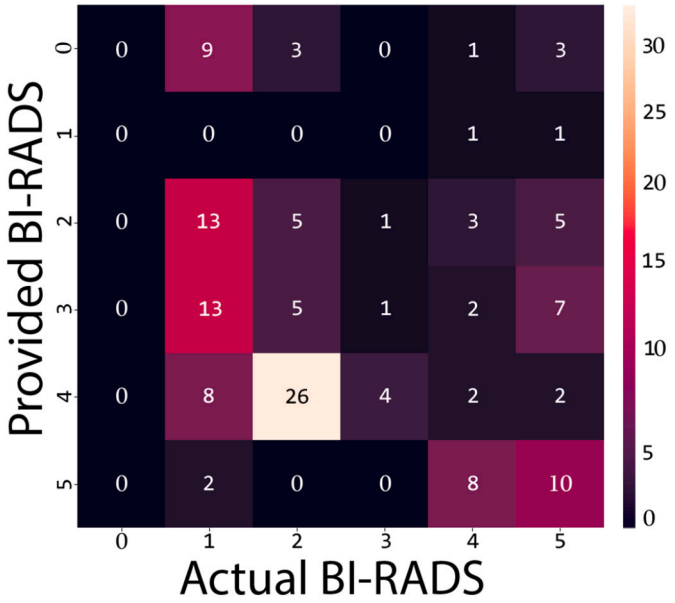


Fig. 5. Confusion Matrix without our AI (*Clinician-Only*) technique. For this case, we did not use our DenseNet results. Each physician directly provided the BI-RADS to the system, with no support of AI techniques. The classification accuracy of clinicians was just 40% for the number of True-Positives. On the other hand, the number of False-Negatives was 6% and the number of False-Positives was 54%. The *Provided* value was most accurate in classifying high (BI-RADS $\geq$ 4) severity cases. The columns represent the *Actual* (biopsy confirmed) category of the objects, and the rows represent the *Provided* (only physician) value. For these percentages, note that we removed several criteria, *e.g.*, BI-RADS = 0, between others.

result. Another factor to take into account is that the BI-RADS is a scale, *i.e.*, it has an order. This means that, if the model says the BI-RADS is 3 (*Medium*), when the real BI-RADS is 5 (*High*), it should be less serious than the model saying it is a BI-RADS of 1 (*Low*).

4. Results

In this section, we experimentally show the usefulness of evaluating the Human-AI behavior, namely its gains when compared to the single radiologist performance. The experimental evaluation will be conducted to map the Research Questions (RQs) formulated in Section 1. The study will comprise the following setups.

We distributed the collaboration (within-subject) into three scenarios: (i) *Clinician-Only*; (ii) *Clinician-AI*; and (iii) *AI-Only*. For each scenario, three patient classes are considered. Concretely, *Low*, *Medium* and *High* severities. Each class concerns the lesion severity in the BI-RADS score, as BIRADS = 1, BIRADS ϵ {2,3}, and BIRADS ϵ {4,5}, respectively.

4.1. RQ1 - clinical impact

Concerning the first research question (RQ1), Figs. 4 and 5 show the metric statistics using the confusion matrix (with 45 clinicians). More precisely, we can observe that the performance accuracy with the proposed AI integration (Fig. 4) is superior in comparison with the performance without integration (Fig. 5). In this confusion matrix, we can denote that the diagonal shape of the matrix is more evident with the introduction of the AI (Fig. 4), in comparison with no AI (Fig. 5).

Improvements of diagnostic accuracy are related with the number of correctly identified cases during the classification (Fig. 4 and Fig. 5) that are always higher in comparison to the results obtained without the assistant. Specifically, we obtained the mean values of (M = 19.2, SD = 12.81) and (M = 3.6, SD = 4.03), with and without the assistance, respectively. The values are taken from the diagonal of the confusion matrices. The diagnostic accuracy is higher (Table 2) when the assistant is integrated in the UI. We obtained[2] the mean and standard deviation of (M = 0.656, SD = 0.34) and (M = 0.622, SD = 0.27) for the *Precision* and *Recall*, respectively. From the above results, the hypothesis **H1.1.** is therefore supported.

Table 2
Reported diagnostic accuracy of Clinician-Only (Clini.-On.) and Clinician-AI (Clini.-AI) collaborations. The *Precision* is the fraction of relevance instances among the classified instances. On the other hand, the *Recall* (sensitivity) is the fraction of relevant instances classified over the total amount of relevant instances.

BIRADS	Precision		Recall	
	Clini.-On.	Clini.-AI	Clini.-On.	Clini.-AI
1	0.01	0.69	0.76	0.91
2	0.13	0.86	0.91	0.91
3	0.20	0.50	0.50	0.20
4	0.20	0.99	0.61	0.68
5	0.99	0.99	0.57	0.86

In terms of time performance, clinicians took 31% less diagnostic time with the assistant (M = 308 s, SD = 57.03 s) in comparison with no assistance (M = 377 s, SD = 44.56 s). Therefore, we can answer the **H1.2.** null-hypothesis, since we achieve improvements of time performance in both FPs and FNs.

In our work, providing the ability to *accept* or *reject* the AI recommendation revealed a significant ($p < 0.05$) positive impact, with a higher feeling of control for clinicians that agree with the competency of the assistance than for those that did not agree. Now analyzing the 14% of clinicians who did *Reject*, we should highlight that from this set, 52% of the clinicians failed (M = 3.72, SD = 4.68) in the classification, contributing to the higher FPs or FNs rates. Thus, we found that the integration of the assistant helps clinicians to improve the final rates, *i.e.*, FPs and FNs (**H1.1.**), as well as time performance (**H1.2.**).

Achieving improvements of time performance in both FPs and FNs means that the assistant is not a distraction for clinicians. Not only we are decreasing the medical errors, but also, successfully telling clinicians where the lesions are. Improving clinicians' time performance without loosing their focus. Hence, we considered the **RQ1** covered.

4.2. RQ2 - user expectations

Concerning the explanation functionality [27] for clinicians (*i.e.*, the XAI of the system), we adopted a visual information technique that relies on the use of *heatmaps* (Fig. 1) to highlight severity regions. The *heatmap* is a way of visualizing the lesion severity, so that the higher values appear closer to red and lower values appear closer to blue. Herein, we take into consideration the following two features for the colored-visualization: (1) the lesion circularity of masses or distribution of microcalcifications; and (2) the associated BI-RADS value computed by the DL model.

It is known that the mass morphology is a relevant feature for the lesion classification [96]. The more irregular the lesion is, the more severe its classification is. Thus, hot colors are associated to more irregular morphology, improving the user's expectation regarding the final result of the AI recommendation. Concerning the BI-RADS, we also assign hot colors for increasing values. This can be easily accomplished by the clinician, by simply press the *6.2. Explain* button.

From the DOTS questionnaire, 98% of the 45 clinicians answered that they do understand what the system is thinking, answering our

[2] *N*: the number of users (Clinicians); F_{var}: the F-test used for comparing the factors of the total deviation per each variable (*var*) categorized by clinical experience; M_{var}: Mean value of the variable (*var*); SD_{var}: the Standard Deviation (SD) per each variable (*var*).

Table 3
Results obtained by computing the Coefficient of Variability (CV) and the respective (a) Inter-Variability (Inter-Var.) and (b) Intra-Variability (Intra-Var.) measurements. The table is divided in Groups of medical experience (*i.e.*, Interns, Juniors, Middles and Seniors) and levels (*i.e.*, Low, Medium and High) of patient severity. Each medical experience is divided in two setups: (i) Clinician-Only (Cli.-Only); and (ii) Clinician-AI (Cli.-AI).

Groups	Low		Medium		High		Intra-Var.	
	Cli.-Only	Cli.-AI	Cli.-Only	Cli.-AI	Cli.-Only	Cli.-AI	Cli.-Only	Cli.-AI
Interns	52.92%	63.83%	11.55%	13.27%	43.32%	10.76%	35.93%	29.28%
Juniors	52.70%	28.40%	29.66%	15.56%	49.49%	15.91%	43.95%	20.29%
Middles	79.70%	45.07%	32.54%	46.19%	46.35%	23.60%	52.87%	38.29%
Seniors	45.46%	48.45%	53.76%	39.37%	56.93%	9.42%	52.05%	32.41%
Inter-Var.	57.69%	46.69%	31.88%	28.60%	49.02%	14.92%	–	–

H2.1. hypothesis. Also, 93% of the clinicians trust the system capability, answering **H2.2.** regarding the confidence of clinicians on the AI recommendation to a proper patient diagnosis. Finally, 91% accept and prefer the AI setup, answering the **H2.3.** hypothesis, as well.

Through the DOTS questionnaire, the following scores for understanding were obtained. Concretely, ($M = 19.7$, $SD = 2.92$) and ($M = 10$, $SD = 7$) when clinicians *press* or *do not press* the *Explain* button, respectively. Furthermore, the values of trust in our assistant were also much higher ($M = 18.81$, $SD = 2.96$) if we compare with clinicians who did not press the *6.2. Explain* button ($M = 12.43$, $SD = 3.21$). Similarly, no significant imbalance was present in prior frequency for the use of *BreastScreening-AI*.

Another important functionality for the clinician's acceptance is to provide the control over the AI recommendation. This can be accomplished by assigning to the *control* with *Accept* or *Reject* functionalities as a way to provide (or inhibit) a second opinion, resulted by the assistant. From DOTS, we obtained 76% of the clinicians that agreed to *Accept* with a positive benevolence statistics ($M = 14.67$, $SD = 5.24$).

Explanation intervention is significantly increasing clinicians' perceived level, both in terms of understanding and trust capability. This suggests that our assistant design is helping to improve user perception by promoting an increase of understanding, trust, and acceptance on AI assistance. That said, we consider the **RQ2** covered.

4.3. RQ3 - Inter-Variability & Intra-Variability

In order to express the clinicians' variability [118] of both *Clinician-Only* and *Clinician-AI* setups, we report (Table 3) two measures of the Coefficient of Variability (CV) as: (a) Inter-Variability; and (b) Intra-Variability. The CV is a measure that is defined as the Standard Deviation (SD) of a measurement set divided by the mean of that same set. The lower the CV, the lower is the dispersion between Inter-Variability and Intra-Variability groups.

In our study, we considered the Inter-Variability (CV_{inter}) as the CV results of all clinicians' set while diagnosing each of the three classified patients, *i.e.*, Low, Medium or High severities. On the other hand, Intra-Variability (CV_{intra}) represents the CV results per (intra) group of medical experience, *i.e.*, Interns, Juniors, Middles, or Seniors. To examine clinicians' variability, we assessed both Inter-Variability (CV_{inter}) and Intra-Variability (CV_{intra}) agreement using percentage of concordance and statistics.

On patients with Low severities, the Inter-Variability improved with the introduction of our assistant ($CV_{inter} = 46.69\%$) in comparison with no AI ($CV_{inter} = 57.69\%$). A total of 11% improvement. For classified patients with Medium severities, the improvement was of 3.28% with the introduction of AI. In terms of classified patients with High severities (which are the critical cases), the improvements were of 34.10%. Therefore, the **H3.1.** null-hypothesis is supported by the results as the Inter-Variability of the diagnosis made by the clinicians improved on patients with Low, Medium and High severities.

For the Intra-Variability, we found that all groups improved their results with the introduction of AI. More precisely, Interns improved on the AI setup ($CV_{inter} = 29.28\%$) by a 6.65% in comparison with no AI ($CV_{inter} = 35.93\%$). From the group of Juniors, the improvements are even higher. With a 23.66% improvement, the variability of Juniors was reduced from a no AI setup ($CV_{intra} = 43.95\%$) to our AI setup ($CV_{intra} = 20.29\%$). On the same hand, Middles reduced their variability by a 14.58%. Finally, Seniors reduced the variability to a 19.64%. Making it evident that the **H3.2.** null-hypothesis is also supported by our results.

Table 4
The *Precision* and *Recall* measurements for the *Clinician-Only* setup. In this case, we grouped severities into *Low*, *Medium* and *High*.

	Precision		Recall	
Severity	Mean	SD	Mean	SD
Low	0.50	0.35	0.63	0.09
Medium	0.17	0.05	0.71	0.05
High	0.60	0.02	0.59	0.14

4.4. Altering clinical workflows

Here, we highlight some results obtained in Section 4 concerning the clinical workflow changes. For our study, three groups of patients were considered: *Low*, *Medium* and *High* severities. As follows, we will detail and correlate each group with the respective obtained results.

4.4.1. Precision & recall

We should highlight that there is a correlation between the above groups and the scores of *Precision* and *Recall*. This is interesting, since the *Clinician-AI* (Table 5) scenario complies with what we observe for the *Clinician-Only* (Table 4) scenario. Specifically, it is easier for the clinician to classify exams with *Low* and *High* severities. Notice that, for these cases, we obtain higher scores for both *Precision* and *Recall* (with smaller standard deviation). This contrasts to the classification performance in the exams that belongs to *Medium* severity (see lower values for the metrics).

The reason behind is that, in the extreme scenarios (*i.e.*, *Low* and *High*) the classification is easier than the *Medium* scenario. Concretely, in *Low* severity, the usual decision is to assign a pathology of "no findings" (no lesion is present), while in *High* severity the lesion is usually of large dimensions and present a specular (irregular shaped) contours, that are quite visible. However, the *Medium* severity is regarded as a "midterm" in the sense that, although the lesion is present, it is not straightforward to classify them as benign or malign. These are the cases where clinicians are making more mistakes. The above issues can be testified in Table 4,

Table 5
The *Precision* and *Recall* measurements for the *Clinician-AI* setup. Again, we grouped into *Low*, *Medium* and *High*.

	Precision		Recall	
Severity	Mean	SD	Mean	SD
Low	0.59	0.07	0.70	0.14
Medium	0.68	0.06	0.56	0.01
High	0.96	0.12	0.77	0.25

where a minimum value for the *Precision* ($M_{\text{Precision}} = 0.17$, $SD_{\text{Precision}} = 0.05$) is obtained for the *Medium* severity.

The great advantage of the proposed assistant, is that we are able to improve the accuracy of the clinician in the most challenging cases (*i.e.*, *Medium* severity) where it is hard to perform a reliable classification. This can be seen in the proposed *Clinician-AI* (Table 5) scenario, where we now obtain a higher score for the precision ($M_{\text{Precision}} = 0.68$, $SD_{\text{Precision}} = 0.06$). Also note that both *Low* and *High* severities present superior performance, whereas the latter exhibits the most notable improvement.

4.4.2. Accuracy & acceptance

Now, regarding the qualitative results, we want to address the most important clinician's opinions and feedback. The feedback and results enable us to conclude that our system achieves a significantly higher expectations of accuracy. Hence, assistant, with the collaboration of clinicians, will likely improve the rates of TPs (Fig. 4). In line with these promising results, the level of benevolence was increased, providing clinicians the opportunity to control the acceptance or rejection of the results. Nevertheless, our system lead clinicians to higher rates of trust and satisfaction by also providing an assistant as a second opinion.

Our study show that a threefold of questions: (1) **Understanding**; (2) **Capability**; and (3) **Benevolence**; can positively answer this respective requirement. Indeed, Cai et al. [27] concluded that the ability to understand what the system is thinking, the integrity of the system to inform the user of being capable for the task, and benevolence are the three factors to explain concisely the user variation in trust. Also, the existence of *heatmaps* were pointed by 82% of the clinicians as an important functionality, promoting higher levels of trust, understanding and acceptance of our assistant on the clinical workflow.

4.4.3. Sensitivity & specificity

Before we can discuss diagnostic precision and recall, we need to know the prevalence of the condition, which is the percentage of the population that has the condition [47]. From literature [23], the estimated prevalence of breast cancer is between 0.75 and 0.90 sensitivity and 0.90 to 0.95 specificity. In our study, we will assume the high end, with sensitivity of 0.90 and specificity of 0.95. In other words, if a woman who has breast cancer gets imaging reports, the test will come back positive 90% of the time because the test has a 0.9 sensitivity. For 10% of the women with breast cancer, the imaging results will be an FN, where the test misdiagnoses a woman who has the disease. Specificity is the accuracy of the diagnosis for women who do not have cancer. That is, if a woman does not have cancer, the test will be negative 95% of the time. The other 5% of the time, we get an FP, which means the test is positive for a woman who does not have breast cancer.

Consistent with speculations from previous works [44,132], the support of **H1.1.** hypothesis from **RQ1.**, and **H2.3.** hypothesis from **RQ2.** shows that clinicians' accuracy and acceptance of a clinical system optimized for high *Precision* can be significantly higher than for a system optimized for higher *Recall* values. For instance (Table 2 and Table 4), taking the *Medium* severity case as an example, we observed that 83% of the clinicians make mistakes, from which 71% are making the FPs (*Recall*). This aspect underlines the need of having more studies regarding *Precision/Recall* understanding impact to the workflow.

Machine Learning (ML) can be a useful tool to match the above requirement on clinical institutions [53,86]. Furthermore, hypotheses **H1.1.**, and **H1.2.** of **RQ1.**, as well as hypotheses **H2.1.**, **H2.2.**, and **H2.3.** of **RQ2.** are confirmed by our results. Measuring FP/FN rates and time performance, suggest that the adjustment of perceptional techniques successfully impact the intended aspects of expectations. Finally, **H3.1.** and **H3.2.** of **RQ3.** were also supported, showing that the techniques are successful in reducing both inter-variability and intra-variability with an AI system.

Our work provides insights into feasible preparation techniques for clinical end-users interacting with medical decision-making systems for a breast cancer diagnosis purpose. This is especially valuable as our assistant techniques are very simple and easy to apply. As a matter of fact, the *BreastScreening* framework is web-based. Therefore, it can be deployed in any remote server and accessed *via* web browser. Which is typically available in any computer device.

4.4.4. Diagnostic time

In our study, clinicians spent about 89 s/image for the **P1** - Low severity, while using our assistant. On the contrary, clinicians spent about 146 s/image, while using a system without any AI technique. Meaning that we have 60% of higher time performance with our assistant, in comparison to a system with no AI. Furthermore, by looking at the time spent on the **P2** - Medium severity, clinicians spent almost the same time. With the assistant the time spent was 77 s/image *vs.* 78 s/image with no AI. Last but not least, for the **P3** - High severity, clinicians did a 116 s/image without AI techniques and with the assistant, clinicians did a 64 s/image. The higher results for the *Low* and *High* results can be easily explained as follows. Results can be easily explained as follows. Results can be easily explained as follows.

In *Low* severity (*i.e.*, no lesion found) even the radiologist does not see any suspicious region, the clinician wants to make sure about this decision. In *High* (*i.e.*, presence of malign lesion), the clinician tends to spend more time to ascertain if there are more lesions to take in consideration. This means that, for an overall of severity conditions, clinicians are performing with better values of time while interacting with our assistant. Not only we achieve, as well as report, better results of *Precision* and *Recall* accuracy. But also, we significantly improve time, accuracy and acceptance.

In our work, we show that clinician's accuracy and acceptance can be improved through the introduction of AI-based assistance for breast cancer diagnosis. Not only in terms of deception [68], but also in terms of the involved understanding for what the system can do. Which is used in typical intelligible AI works [1], and can be applied to this type of medical decision-making systems. At the end, this addresses an important gap in existing research on preparing clinicians for AI. Providing a new paradigm shift, by assisting clinicians and patient treatment in a more efficient way.

4.4.5. Clinical impact

We found that our expectation adjustments offer significant improvement in accuracy and acceptance across higher levels of *Recall*. Moreover, those adjustments proved effective high *Precision* and *Recall* rates in which clinicians experienced higher levels of trust. We believe this shows that expectation adjustments are intended to clinicians. As explained in our work, we developed the study to expose clinicians to our AI-assisted techniques in order to check if the preparation through expectation adjustments can be an effective approach on the clinical domain.

Few studies have been performed to evaluate how a CDSS can be implemented in a manner that maximizes their clinical impact, which must be the ultimate target. Even with the presented limitations, it is expected that HAII will play a major role in the evaluation of those systems [16,17]. In the *BreastScreening-AI* framework, having a method that has high specificity to minimize the FP outcomes may lead to unnecessary anxiety among patients and negatively impact the cost-effectiveness.

As we presented in our results section, the analysis of significance for high severity cases (BI-RADS > 3) shows that our assistant will impact on the radiologists choice of malignant lesions. The results proved a 51% accuracy improvements of these high severity cases during the use of our AI-assisted system. With this achieved improvements, it is trivial to understand that AI will positively impact the clinical domain.

In short, AI is progressively having a cognitive impact in medical practice by applying several AI-assisted techniques (*i.e.*, DenseNet [62]) to rapidly and automatically read the medical images. As above described, AI can provide new paths to optimize the healthcare using

medical imaging, and may provide new therapies, reducing medical errors and clinical trials analysis [10].

4.4.6. Perception differences

Directly from medical data, AI can advert clinical errors due to cognitive biases, positively impacting patient care. In this study, we show that, despite several achieved tasks in which the AI functionalities are supposed to assist, clinicians should circumspect the result decisions for the impact of various types of assistant flaws. Particularly, for the impact of various types of assistant flaws such as *workload*, *usability* and *usefulness*.

For *workload*, we measure both *mental* and *physical* demands. Besides, *mental* measurements are bringing us information regarding how do clinicians perceive while ignoring incorrect suggestions and scanning multiple suggestions. Contrary, *physical* measurements provide us information regarding how do they perceive while having to execute a new task manually or reverse an incorrect assistant action.

In our work, we focus on the *Precision* and *Recall* problem, since we want to measure the perception differences of a decision-making system for medical purposes. That said, we do not show on the results section the other acquired metrics, such as *workload*, that we address now. To measure *workload*, we used the well known NASA-TLX [63,117] scale. For the statistical significance, we used the ANOVA statistical tests [99,138]. Briefly, both *mental* and *physical* measurements perceived lower (which is better) with our assistant ($F = 2.85$) in comparison to the condition without the AI technique ($F = 7.86$). Both results showed to have a significant impact ($p < 0.05$) for the two (*i.e.*, *mental* and *physical*) *workload* measurements.

For the *usability*, we measure it in terms of performance and learnability, as well as satisfaction. Performance denotes the accuracy and completeness of clinicians in accomplishing the specified system functionalities. Learnability is one of the measurements for effectiveness, and it assists with abilities in functioning the system. Satisfaction indicates the insights and opinions of the system. To measure *usability*, we used the SUS scale [63,117]. In short, 86% of clinicians agree that the introduction of AI assistance will not bring more complexity to the diagnosis task [9]. Moreover, 85% of clinicians preferred the assistant condition against the opposite. Meaning that the overall *usability* results are positive.

Finally, we also measure *usefulness*. An early qualitative analysis showed us that (41/45) clinicians accept the introduction of an AI algorithm to their daily work. In fact, many of the clinicians said that: “I would like to frequently use your system on my daily practice, as it can improve my performance” (C1). Which is representative for the acceptance of the system, making the assistant a valuable tool. Additionally, others said that: “AI will change the way radiology is performed. New quantitative information will be added to reports and repetitive tasks will be made automatically” (C15). In many ways, clinicians were making sense of this new technology shift in terms of their own clinical practices. Indeed, comments such as “AI will be in the future, the role, and workflow to be determined” (C7) are showing less resistance of change. As capabilities of AI are reaching human performance levels, considering AI through the lens of existing clinical practices may become increasingly pertinent.

To conclude this section, with AI we can advert clinical errors by taking advantage of our medical data. At a same time, we improve not only patient care, but also several other aspects for clinicians. From accuracy of the diagnostic results, to the reduction of clinicians' *workload*. With the introduction of our AI-assisted techniques, we improved several perception differences of the cognitive biases so present on the medical decision-making systems [6].

5. Discussion

In this work, we studied acceptance and receptiveness of clinicians, as well as how perceived UX can affect the introduction of *AI-Assisted* medical imaging systems into the clinical workflow. We conducted a within-subject study to investigate the use of AI assistants by medical experts. Our results show that AI assistants can alter medical workflows by improving clinicians' performance, as well as acceptance of an AI-assisted system, while maintaining overall diagnostic productivity. Moreover, clinicians' overall performance was not affected by providing control or explainable techniques over the final diagnostic. Indeed, we observed a significant effect of decreasing the medical error (*i.e.*, rates of FPs and FNs) on clinicians' case correction rate. This promising insight motivates future research into the development and validation of AI assistants capable of providing highly relevant explanations as a second reader diagnostic tool.

5.1. Findings

BreastScreening-AI represents a type of systems that is passive (*i.e.*, assistive) and casual. Passive, since clinicians can *Accept*, *Reject* and ask the system to *Explain*, or even *ignore* the *Assistant* result. The causality results from the training strategy used for the DenseNet. The output classification (present information) of this network always depend on the images that the network received previously (past information). While this is only one class of CDSS, we believe this class of systems represents many current efforts of integrating HAII into other clinical domains (*e.g.*, systems for clinical drug development, epidemiology, dementia treatment) [124,129,132].

Hence, our findings should generalize to other clinical systems and tasks on a threefold of levels:

1. We reveal whether clinicians need explanations during the medical imaging diagnosis;
2. We elicit system requirements that encompass the current practices and the future of AI-assisted diagnostic;
3. We provide evidence detailing the benefits of avoiding the different types of clinical errors;
4. We suggest how to optimize both *Precision* and *Recall*, reducing FPs and FNs, respectively;

Regarding (1), it is known that in critical systems, it is more important to analyze the severity of consequences. However, the severity of consequences applies to different clinical decision-making (*i.e.*, the classification) doubts rather than any medical imaging workload improvements (*e.g.*, drug development, epidemiology, or dementia). One of the major doubts and difficulties came from the *Medium* cases, which must be a higher concern regarding explanations. To accomplish this, when the clinician presses the *Explain* button, the information showed by the *heatmaps* must further consider those cases.

In (2), by comparing the clinicians' behavior with respect to *Clinician-Only vs. Clinician-AI*, we obtain system requirements, that encompass the current practices and the future of AI-assisted diagnosis. One important requirement, was to provide clinician control over the proposed BI-RADS resulted from the AI algorithm. With the *Accept* and *Reject* buttons, the clinician can control the final diagnostic value. In case of *Reject*, the clinician can even input and provide the new BI-RADS value.

In (3), the importance of avoiding the different type of errors depends on the information available from a specific clinical domain. Indeed, this is a complex issue. While we could argue that for the breast cancer diagnosis FNs are always better to avoid, we can not always suggest that independently regrading the clinical domain. In our case, missing FNs has more impact than marking an FP.

Finally, in (4), from other clinical domains, an FP might be more important to avoid. For instance, on drug development, we want to prefer avoiding the FP rates. By doing that, we are optimizing the number of chemicals and drugs identified incorrectly by the system [116]. That said, we believe our findings generalizes to a clinical class of passive systems (*i.e.*, clinicians are making the final decisions) in which the end ratio of workload regarding both FPs and FNs is lower.

5.2. Further work

Breast cancer screening is one of the nuclear topics in medical imaging analysis. We believe that with our proposed assistant, we are able to change diagnosis paradigm. Our results indicate that it is possible to introduce behavioral change on the clinical workflow in a HAII manner [14]. Further work will include other deep network architectures, besides the DenseNet used herein, and to generate attention maps as an alternative to the heatmaps. That is, we will include not only the classification output (as proposed in this paper) but also, the segmentation map of the lesion.

In this paper, the study was performed with a *dataset* of images from one institution. Could be interesting to enlarge the clinician number from this institution, and also enlarging the data from other institutions. This is interesting, since in the recent years, many ML models have been proposed to accurately diagnose breast cancer. However, when these models are tested on unseen datasets, acquired with different devices and scanners, their performance is usually reduced. Thus, there exist a need to build generalizable models that can be applied consistently across clinical centers.

6. Conclusions

In our work, we developed *BreastScreening-AI*, an AI-assisted clinical imaging support system which provides automated diagnosis based on Deep Learning methods. We used *BreastScreening-AI* in a real-world scenario with 45 clinicians from nine clinical institutions to experiment with several techniques of Human-AI Interactions. Our results show improvements in clinician's accuracy, as well as acceptance of an AI-assisted system on the clinical domain for the breast cancer diagnosis. Also, we show that by focusing on both *Precision* and *Recall* problem, the higher rates of the system will perform at the same level of accuracy, leading to much higher perceptions of accuracy. Hence, increasing acceptance of the system by the medical professionals. Before AI-assisted techniques can be used in medicine, we first need to go through clinical trials, so that we can establish our system validity. We need to make sure that the model actually provides benefit and guarantee the value to clinicians, as well as to the final patient. Advances in precision medicine will bring more diagnostic tests and information with more treatment options. Which will only increase the workload of clinicians and decrease of expectations. Indeed, the introduction of AI systems in the clinical workflow will change medical interventions. Our findings open the way of shaping expectations among medical decision-making as an effective way of improving clinician's acceptance of AI systems on the clinical domain.

7. Neural network integration

In this section, we provide additional details concerning the use of the DenseNet that is integrated in our UI. Concretely, we provide additional details concerning: (i) the dataset; (ii) data pre-processing; (iii) evaluation strategy; (iv) learning methodology; (v) transfer learning; and (vi) additional description of the system.

7.1. Comparison with existing methodologies

In terms of model comparison with already existing methods, the literature shows improvements of the DenseNet model concerning mean sensitivity and specificity by surpassing, as an example, the EfficientNet model [131]. For instance, the authors report the Mean AUC for detecting breast cancer was 0.952 ± 0.005 by a DenseNet model and 0.954 ± 0.020 by an EfficientNet model. Moreover, the authors further report the mean sensitivity and specificity from a DenseNet model, of 87% and 88%, respectively, surpassed the mean values, of 81% and 82%, respectively, obtained in their meta-analysis. Hence, the literature has demonstrated the DenseNet model efficiency [30,67,143].

7.2. Dataset size

In this paper, we used a total of 338 cases and acquired in the Hospital Fernando Fonseca. From this set of cases, 289 were classified by the head of radiology. Each patient has several images concerning four X-ray mammography (two in CC and two MLO views), one ultrasound image, and roughly 5 images in MRI. In the MRI volumes, we take several image slices per patient, where the lesion is present (recall point 5) in the previous question). This provides us roughly 2890 images (*i.e.*, $289 \times (4 + 1 + 5)$), that is used to train/test the AI agent (DenseNet).

7.3. Data pre-processing

Traditional image processing and deep learning techniques require extensive pre-processing [143]. As a matter of fact, it is known that a cleaning dataset is welcome when training a deep neural network [119]. In our study this stage is of the utmost importance, since the MG, US and MR images contain quite different intensities and the images are of different sizes. Thus, before introducing the images to the DenseNet, we pre-process the data. Specifically, we perform a data normalization, so that the images have the same intensity, regardless of the modality.

All the images are resized to 224×224 pixels. Also, the images are normalized by subtracting its mean and dividing by its standard deviation. The above size is used since the DenseNet is prepared to receive the input with this format.

7.4. Training and evaluation of the DenseNet

The evaluation mechanism for the DenseNet follows a two stage process, involving a training and testing stages. To accomplish this, we partition the dataset (see Section 7.2) to form the training and testing subsets in the following manner: 80% of the samples are used for training and the remaining 20% for testing. In this partition, we guarantee that each set (training and test) is properly balanced. That is, each set contains samples that belong to all five and in all modalities (*i.e.*, MG, US and MRI) - this is to guarantee the multimodality.

Training: The training follows a supervised learning strategy. Specifically, during the training we provide the image, as well as the corresponding label, that is, the classification BI-RADS score. This classification (*i.e.*, ground truth) is provided by the head of radiology of the HFF institution. This allows the DenseNet the ability to establish associations between the morphology of the lesion (image) and its severity (classification label).

Testing: During the test phase, we used a hold-out test set that aimed to measure the performance of the DenseNet. An accuracy of 94.81% was obtained in this test set. Notice that the three patients (as mentioned in Sec 4.4) are obtained from this hold-out test set.

Remark: Another possibility for the DenseNet training is to resort to unsupervised learning methods. However, this class of approaches is more challenging, since only the images are fed into the network without the help of having the classification labels (*i.e.*, without BI-RADS classification in training). This deserves our attention in a future work.

7.5. Transfer learning

As a final remark, we should stress that although the number of samples is relatively small (3000 as mentioned in Section 7.2), we may argue that this could be not enough, since a robust training DNNs usually requires large amounts of annotated samples (order of 100Ks) to avoid overfitting to the training data given the large model capacity. This issue can be handled with the so-called “Transfer Learning” (TL). Basically, the TL process retrains (in a process called fine-tuning) publicly available models (pre-trained with large datasets) using smaller datasets.

To initialize the weights of the network, the DenseNet model was pre-trained on the ImageNet dataset [48] with more than 1.2 million images, which yields fast convergence for deep architectures [51]. We

Table 6
Evaluation results of the AI-Only for the unimodal tests with the DenseNet-121 model. Compared Models as unimodal training: Cranio-Caudal MG (CC-MG); Medio-Lateral Oblique MG (MLO-MG); UltraSound (US); and Dynamic Contrast-Enhanced MRI (DCE-MRI). Finaly, the multimodal training stands for CC-MG, MLO-MG, US, and DCE-MRI all modalities together.

Strategy	Unimodal				Multimodal
Modalities	CC-MG	MLO-MG	US	DCE-MRI	
Accuracy	0.73	0.68	0.65	0.667	0.9481

used the DenseNet model by following two particular literature works for the classification of mammograms [80,121].

It has been demonstrated that this TL approach is possible to be applied in the medical image field, even if the datasets used in this pre-training stage are completely different from the medical images - in this case - different from breast images [40,41]. This is precisely what we did here. Concretely, we used a pre-trained that provides the DNN more regularization and generalization capabilities to new datasets. In the experiments, the DenseNet was pre-trained with the ImageNet model that contains millions of images from 1000 classes (Section 3.5). The only adaptation required to adapt the DenseNet, is to change the last layer from 1000 nodes (or classes) to the number of five classes in our problem.

7.6. DenseNet final remarks

The DenseNet is a complex model. This means that the model can handle heterogeneous types of the input. Thus, given the relatively small number of training samples, it is advantageous to join all the modalities.

In the experiments, we have compared two situations:

1. Joining all the modalities to a single DenseNet; and
2. Using three different DenseNet, each network for each image modality type (MG, US and MRI).

The best results were achieved for situation (1), and this was adopted for our UI.

7.7. Additional description of the system

Now we detail a little better the description of the system, as follows:

1. Each of the 289 patients was classified by the head of radiology of the Medical Imaging Department at the HFF institution. This classification corresponds to assigning a BI-RADS value for each modality image of the exam;
2. We trained our DenseNet across the BI-RADS, (we already detailed how the training/test of the DenseNet is accomplished);
3. After the training, we have the test phase. Here the new (unseen) images are presented to the DenseNet, and a BI-RADS classification is provided to that test sample; and
4. To evaluate the accuracy of the system, the precision and recall are computed.

The following link is available that illustrates eases the description of the proposed system, as well as its workflow:
youtu.be/k2vBpbKqJqQ

7.8. Separate vs. all modalities

As shown in Table 6, the performance in the multimodal scenario is significantly higher than in the unimodal case. This is a consequence of substantially increasing the training dataset when using all imaging modalities, which means the model is trained with a much larger number of images and with higher variability. This increase in data helps regularize the DenseNet model, preventing over-fitting and improving its generalization capabilities. In the unimodal case, the lack of a large training dataset resulted in severe over-fitting issues, despite our efforts to mitigate this through: (i) data augmentation strategies, where each image is randomly modified in each batch using common transformations, such as small rotations, flipping, and translations; and

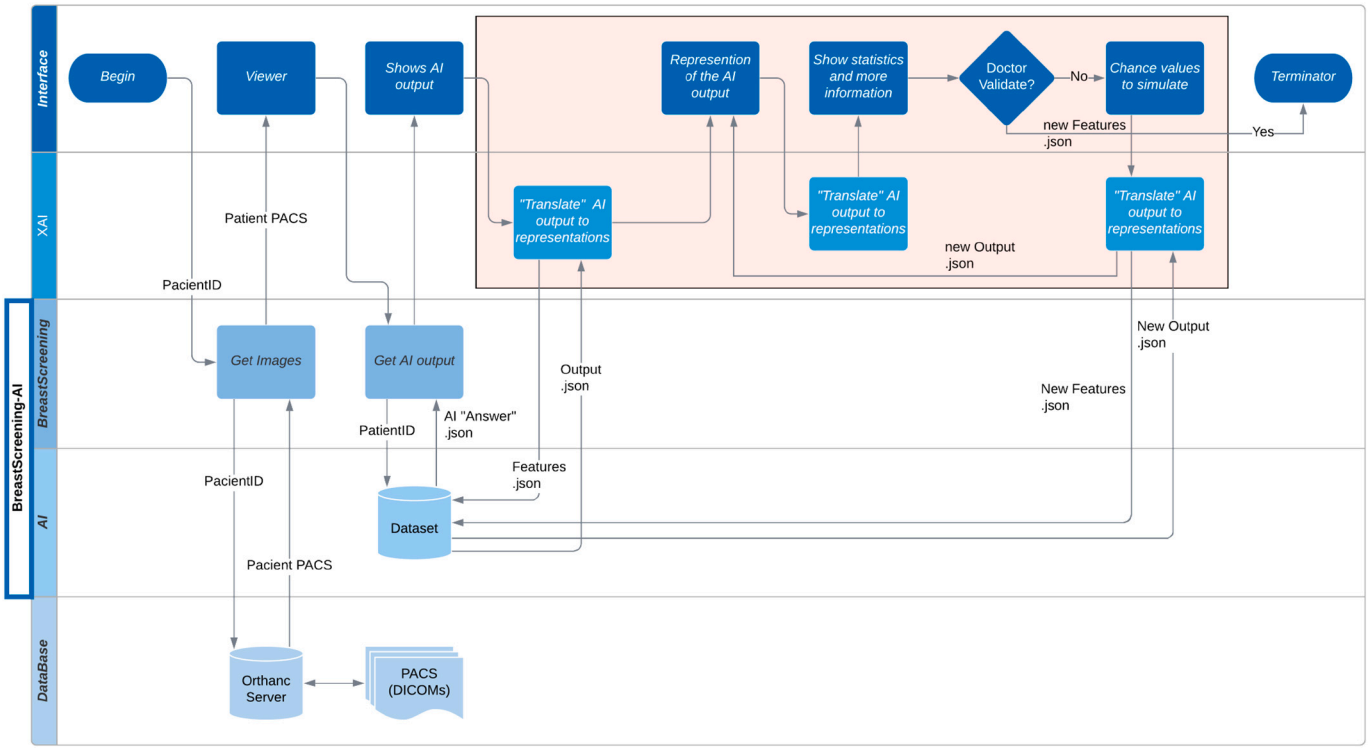


Fig. 6. Architecture and functionalities diagram [108] of the typical use cases. These use cases are presented with the order of actions for the interrogated AI system.

(ii) other regularization strategies like dropout and L2 (weight decay).

As a final remark, we stress that the network architecture receives 2D images. Since the DCE-MRI are volumes, we extract some of the slices that contain the lesion. All the above issues led the network to obtain significantly better results in the multimodal scenario, as demonstrated in Table 6.

7.9. System diagram

Concerning the system diagram [105] of the BreastScreening-AI framework (Fig. 6), the first action is to request the patients (Section 2.3) stored on the Orthanc server [81]. Next, the assistant will show the AI outputs [34,108]. Finally, clinicians can decide by accepting or rejecting the AI suggestion.

As stated before (Section 2), our interface (Fig. 1) has three main button functionalities (*i.e.*, 6.1. Accept; 6.2. Explain; and 6.3. Reject). Each button of our interface is generating different triggers [145] on the framework (Fig. 6). Hence, three triggers are important to denote.

The first trigger is made by clinicians when pressing the accept button, where a JSON file is generated to provide success information to the model [33]. The second trigger represents the AI model explanations. Clinicians press the explain button, in order of having more reliable information concerning what regions were considered by the model and how severe are these lesions to the model. Finally, the third trigger happens when clinicians are rejecting the AI result. At this point, clinicians must provide the new BI-RADS so that we can further retrain the AI model. To this end, the framework will generate a JSON file with the new rejected BI-RADS values provided by clinicians.

7.10. Repositories

For more details about the source code, please follow the `prototype-multi-modality-assistant` (git.io/JMjDi) repository. In terms of DICOM files, the dataset of patients is available in the dataset-uta7-dicom (https://git.io/JMjy7) repository. These links were accessed on 7th of December 2021.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests:

This work was founded by a research grant from Fundação para a Ciência e a Tecnologia (FCT), with the UID/EEA/50009/2013 reference.

Acknowledgements

This work was partially supported by national funds of two Portuguese scientific institutions. Specifically, Fundação para a Ciência e a Tecnologia and Instituto Superior Técnico. The work was supported under the UID/EEA/50009/2013 project and BL89/2017-IST-ID grant.

We would like to thank Dr. Clara Aleluia and her radiology team of Hospital Fernando Fonseca for valuable insights and helping to use the AI assistant on their daily basis. From IPO Lisboa, we would like to thank the medical imaging teams of Dr. José Carlos Marques and Dr. José Venâncio. From IPO Coimbra, we would like to thank the radiology department director and all team of Dr. Idílio Gomes. Moreover, we would like to provide our acknowledgements to Dr. Emília Vieira and Dr. Cátia Pedro from Hospital Santa Maria. We also want to thank all teams of the radiology department of Hospital do Barreiro for participation, as well as from all other clinical institutions, with a special gratitude to Dr. João Maria Abrantes for promoting this collaboration. A great thank to Dr. Cristina Ribeiro da Fonseca who, among others, is giving us crucial information for the *BreastScreening* research work. Finally, we would like to thank our MSc Students, Hugo Lencastre, Nádia Mourão, João Bernardo, Madalena Pedreira, Margarida Morais, Pedro Diogo, Mauro Machado, and João Fernandes for their development, support, feedback and opinions.

References

[1] Abdul A, Vermeulen J, Wang D, Lim BY, Kankanhalli M. Trends and trajectories for explainable, accountable and intelligible systems: an hci research agenda. In: Proceedings of the 2018 CHI Conference on Human Factors in Computing Systems. New York, NY, USA: ACM; 2018. 582:1–582:18. https://doi.org/10.1145/3173574.3174156. URL: http://doi.acm.org/10.1145/3173574.3174156.

[2] Abdul A, von der Weth C, Kankanhalli M, Lim BY. Cogam: measuring and moderating cognitive load in machine learning model explanations. In: Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems. New York, NY, USA: Association for Computing Machinery; 2020. p. 1–14. https://doi.org/10.1145/3313831.3376615. URL: 10.1145/3313831.3376615.

[3] Admoni H. Nonverbal communication in socially assistive human-robot interaction. AI Matters 2016;2:9–10. https://doi.org/10.1145/3008665.3008669. URL: 10.1145/3008665.3008669.

[4] Aha D, Coman A. The ai rebellion: changing the narrative. In: Proceedings of the AAAI Conference on Artificial Intelligence. 31; 2017. URL: https://ojs.aaai.org/index.php/AAAI/article/view/11141.

[5] Aha DW, Coman A. The ai rebellion: changing the narrative. In: Thirty-first AAAI Conference onArtificial Intelligence; 2017. p. 1–5.

[6] Alaimi M, Law E, Pantasdo KD, Oudeyer PY, Sauzeon H. Pedagogical agents for fostering question-asking skills in children. In: Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems. New York, NY, USA: Association for Computing Machinery; 2020. p. 1–13. https://doi.org/10.1145/3313831.3376776. URL: 10.1145/3313831.3376776.

[7] Allen JF, Byron DK, Dzikovska M, Ferguson G, Galescu L, Stent A. Toward conversational human-computer interaction. In: AI Magazine. 22; 2001. p. 27. https://doi.org/10.1609/aimag.v22i4.1590. URL: https://ojs.aaai.org/index.php/aimagazine/article/view/1590.

[8] Amershi S, Weld D, Vorvoreanu M, Fourney A, Nushi B, Collisson P, Suh J, Iqbal S, Bennett PN, Inkpen K, Teevan J, Kikin-Gil R, Horvitz E. Guidelines for human-ai interaction. In: Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems. New York, NY, USA: Association for Computing Machinery; 2019. p. 1–13. https://doi.org/10.1145/3290605.3300233. URL: 10.1145/3290605.3300233.

[9] Amir O, Grosz BJ, Gajos KZ. Mip-nets: enabling information sharing in loosely-coupled teamwork. In: Proceedings of the Thirtieth AAAI Conference on Artificial Intelligence. AAAI Press; 2016. p. 4192–3. URL: https://www.sciencedirect.com/science/article/pii/S0933365719306712.

[10] Amir O, Grosz BJ, Gajos KZ, Swenson SM, Sanders LM. From care plans to care coordination: opportunities for computer support of teamwork in complex healthcare. In: Proceedings of the 33rd Annual ACM Conference on Human Factors in Computing Systems. New York, NY, USA: Association for Computing Machinery; 2015. p. 1419–28. https://doi.org/10.1145/2702123.2702320. URL: 10.1145/2702123.2702320.

[11] Amir O, Grosz BJ, Law E, Stern R. Collaborative health care plan support. In: Proceedings of the 2013 International Conference on Autonomous Agents and Multi-Agent Systems. Richland, SC: International Foundation for Autonomous Agents and Multiagent Systems; 2013. p. 793–6.

[12] Anacleto JC, Fels SS. Lessons from ict design of a healthcare worker-centered system for a chronic mental care hospital. In: CHI '14 Extended Abstracts on Human Factors in Computing Systems. New York, NY, USA: Association for Computing Machinery; 2014. p. 1687–92. https://doi.org/10.1145/2559206.2581260. URL: doi:10.1145/2559206.2581260.

[13] Anik AI, Bunt A. Data-centric explanations: explaining training data of machine learning systems to promote transparency. In: Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems. New York, NY, USA: Association for Computing Machinery; 2021. p. 1–13. https://doi.org/10.1145/3411764.3445736. URL: 10.1145/3411764.3445736.

[14] Arnulphi C, Sánchez SA, Alejandra Tricerri M, Gratton E, Jonas A. Interaction of human apolipoprotein a-i with model membranes exhibiting lipid domains. Biophys J 2005;89:285–95. https://doi.org/10.1529/biophysj.104.047480. URL: https://www.sciencedirect.com/science/article/pii/S0006349505726783.

[15] Azuaje F. Artificial intelligence for precision oncology: beyond patient stratification. NPJ PrecisOncol 2019;3:6.

[16] Bansal G, Nushi B, Kamar E, Lasecki WS, Weld DS, Horvitz E. Beyond accuracy: the role of mental models in human-ai team performance. URL https://ojs.aaai.org/index.php/HCOMP/article/view/5285. In: Proceedings of the AAAI Conference on Human Computation and Crowdsourcing. 7; 2019. p. 2–11.

[17] Bansal G, Nushi B, Kamar E, Weld DS, Lasecki WS, Horvitz E. Updates in human-ai teams: understanding and addressing the performance/compatibility tradeoff. In: Proceedings of the AAAI Conference on Artificial Intelligence. 33; 2019. p. 2429–37. https://doi.org/10.1609/aaai.v33i01.33012429. URL: https://ojs.aaai.org/index.php/AAAI/article/view/4087.

[18] Bansal G, Wu T, Zhou J, Fok R, Nushi B, Kamar E, Ribeiro MT, Weld D. Does the whole exceed its parts? The effect of ai explanations on complementary team performance. In: Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems. New York, NY, USA: Association for Computing Machinery; 2021. p. 1–16. https://doi.org/10.1145/3411764.3445717. URL: 10.1145/3411764.3445717.

[19] Benrimoh D, Fratila R, Israel S, Perlman K, Mirchi N, Desai S, Rosenfeld A, Knappe S, Behrmann J, Rollins C, et al. Aifred health, a deep learning powered clinical decision support system for mental health. In: The NIPS'17 Competition: Building Intelligent Systems. Springer; 2018. p. 251–87.

[20] Bharadhwaj H. Explainable recommender system that maximizes exploration. In: Proceedings of the 24th International Conference on Intelligent User Interfaces: Companion. New York, NY, USA: ACM; 2019. p. 1–2. https://doi.org/10.1145/3308557.3308699. URL: http://doi.acm.org/10.1145/3308557.3308699.

[21] Bhattacharya S, Sharma N, Singh A. Designing culturally acceptable screening for breast cancer through artificial intelligence-two case studies. JFamMedPrimary Care 2019;8:760–2. https://doi.org/10.4103/jfmpc.jfmpc_391_18.

[22] Boughey JC, Ballman KV, Le-Petross HT, McCall LM, Mittendorf EA, Ahrendt GM, Wilke LG, Taback B, Feliberti EC, Hunt KK. Identification and resection of the clipped node decreases the false negative rate of sentinel lymph node surgery in patients presenting with node positive breast cancer (t0–t4, n1–2) who receive neoadjuvant chemotherapy–results from acosog z1071 (alliance). Ann Surg 2016;263:802.

[23] Bray F, Ren JS, Masuyer E, Ferlay J. Global estimates of cancer prevalence for 27 sites in the adult population in 2008. Int J Cancer 2013;132:1133–45. https://doi.org/10.1002/ijc.27711.

[24] Burr C, Cristianini N, Ladyman J. An analysis of the interaction between intelligent software agents and human users. MindsMach 2018;28:735–74. https://doi.org/10.1007/s11023-018-9479-0.

[25] Cadario R, Longoni C, Morewedge CK. Towards trustable machine learning. <sb:contribution><sb:title>Nat </sb:title></sb:contribution><sb:host><sb:issue><sb:series><sb:title>Biomed Eng</sb:title></sb:series></sb:issue></sb:host> 2018;2:709–10. https://doi.org/10.1038/s41551-018-0315-x. URL: 10.1038/s41551-018-0315-x.

[26] Cadario R, Longoni C, Morewedge CK. Understanding, explaining, and utilizing medical artificial intelligence. <sb:contribution><sb:title>Nat Hum</sb:title></sb:contribution><sb:host><sb:issue><sb:series><sb:title>Behav</sb:title></sb:series></sb:issue></sb:host> 2021;5:1636–42. https://doi.org/10.1038/s41562-021-01146-0. URL: 10.1038/s41562-021-01146-0.

[27] Cai CJ, Jongejan J, Holbrook J. The effects of example-based explanations in a machine learning interface. In: Proceedings of the 24th International Conference on Intelligent User Interfaces. New York, NY, USA: Association for Computing Machinery; 2019. p. 258–62. https://doi.org/10.1145/3301275.3302289.

[28] Cai CJ, Reif E, Hegde N, Hipp J, Kim B, Smilkov D, Wattenberg M, Viegas F, Corrado GS, Stumpe MC, Terry M. Human-centered tools for coping with imperfect algorithms during medical decision-making. In: Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems. New York, NY, USA: ACM; 2019. 4:1–4:14. https://doi.org/10.1145/3290605.3300234.

[29] Cai CJ, Winter S, Steiner D, Wilcox L, Terry M. "hello ai": uncovering the onboarding needs of medical practitioners for human-ai collaborative decision-making. ProcACM HumComput Interact 2019;3. https://doi.org/10.1145/3359206.

[30] Cai Q, Liu X, Guo Z. Identifying architectural distortion in mammogram images via a se-densenet model and twice transfer learning. In: 2018 11th International Congress on Image and Signal Processing, BioMedical Engineering and Informatics (CISP-BMEI); 2018. p. 1–6. https://doi.org/10.1109/CISP-BMEI.2018.8633197.

[31] Calisto FM. Medical imaging multimodality breast cancer diagnosis user interface. Avenida Rovisco Pais 1, 1049-001 Lisboa - Portugal (EU): Instituto Superior Técnico; 2017. https://doi.org/10.13140/RG.2.2.15187.02084. URL: http://doi.acm.org/10.1145/3173574.3174156.

[32] Calisto FM. Assistant introduction: user testing guide for a comparison between multi-modality and AI-assisted systems. Technical Report. Instituto Superior Técnico; 2019. https://doi.org/10.13140/RG.2.2.16566.14403/1. URL: http://rgdoi.net/10.13140/RG.2.2.16566.14403/1.

[33] Calisto FM. Breast cancer medical imaging multimodality lesion contours annotating method. Technical Report 116801. URL: http://rgdoi.net/10.13140/RG.2.2.14792.55049. Avenida Rovisco Pais 1, 1049-001 Lisboa - Portugal (EU): Instituto Superior Técnico; 2020. https://doi.org/10.13140/RG.2.2.14792.55049. method and process using a system to annotate and visualize masses and microcalcifications of breast cancer lesions in a multimodality strategy.

[34] Calisto FM. Towards the human-centered design of intelligent agents in medical imaging diagnosis. URL: https://doi.org/10.13140/RG.2.2.12519.16803. Avenida Rovisco Pais 1, 1049-001 Lisboa - Portugal (EU): Instituto Superior Técnico; 2020. https://doi.org/10.13140/RG.2.2.12519.16803. introduction of intelligent agents for breast cancer diagnosis.

[35] Calisto FM, Ferreira A, Nascimento JC, Gonçalves D. Towards touch-based medical image diagnosis annotation. In: Int'l Conf. Interactive Surfaces and Spaces (ISS). New York, NY, USA: ACM; 2017. p. 390–5. https://doi.org/10.1145/3132272.3134111.

[36] Calisto FM, Nunes N, Nascimento JC. Breastscreening: on the use of multi-modality in medical imaging diagnosis. In: Proceedings of the International Conference on Advanced Visual Interfaces. New York, NY, USA: Association for Computing Machinery; 2020. p. 1–5. https://doi.org/10.1145/3399715.3399744.

[37] Calisto FM, Santiago C, Nunes N, Nascimento JC. Introduction of human-centric ai assistant to aid radiologists for multimodal breast image classification. IntJHumComputStud 2021;150:102607. https://doi.org/10.1016/j.ijhcs.2021.102607.

[38] Callaghan W, Goh J, Mohareb M, Lim A, Law E. Mechanicalheart: a human-machine framework for the classification of phonocardiograms. ProcACM HumComput Interact 2018;2. https://doi.org/10.1145/3274297.

[39] Cardoso MJ, Houssami N, Pozzi G, Séroussi B. Artificial intelligence (ai) in breast cancer care - leveraging multidisciplinary skills to improve care. Technical Report Artif Intell Med 2020:102000. https://doi.org/10.1016/j.artmed.2020.102000. URL: http://rgdoi.net/10.13140/RG.2.2.16566.14403/1.

[40] Carneiro G, Nascimento J, Bradley AP. Unregistered multiview mammogram analysis with pre-trained deep learning models. In: Navab N, Hornegger J, Wells WM, Frangi AF, editors. Medical image computing and computer-assisted intervention –. Cham: Springer International Publishing; 2015. p. 652–60. https://doi.org/10.1007/978-3-319-24574-4_78.

[41] Carneiro G, Nascimento J, Bradley AP. Automated analysis of unregistered multi-view mammograms with deep learning. IEEE Trans Med Imaging 2017;36:2355–65. https://doi.org/10.1109/TMI.2017.2751523.

[42] Chatelain P, Sharma H, Drukker L, Papageorghiou AT, Noble JA. Evaluation of gaze tracking calibration for longitudinal biomedical imaging studies. IEEE TransCybern 2018:1–11.

[43] Chattopadhyay P, Yadav D, Prabhu V, Chandrasekaran A, Das A, Lee S, Batra D, Parikh D. Evaluating visual conversational agents via cooperative human-ai games. In: Fifth AAAI Conference on Human Computation and Crowdsourcing; 2017. p. 1–9.

[44] Chaurasia V, Pal S. A novel approach for breast cancer detection using data mining techniques. Int J Innov Res Comput Commun Eng 2017;2. URL: https://www.jmir.org/2022/1/e28659.

[45] Chen X, Williams BM, Vallabhaneni SR, Czanner G, Williams R, Zheng Y. Learning active contour models for medical image segmentation. In: Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition; 2019. p. 11632–40. https://doi.org/10.1109/CVPR.2019.01190.

[46] Chin-Yee B, Upshur R. Clinical judgement in the era of big data and predictive analytics. J Eval Clin Pract 2018;24:638–45.

[47] Chkotua S, Peleteiro B. Peer reviewed: mammography use in Portugal: national health survey 2014. Prev Chronic Dis 2017;14:1–2. https://doi.org/10.5888/pcd14.170054.

[48] Deng J, Dong W, Socher R, Li LJ, Li K, Fei-Fei L. Imagenet: a large-scale hierarchical image database. In: 2009 IEEE conference on computer vision and pattern recognition. IEEE; 2009. p. 248–55.

[49] Depeursinge A, Duc S, Eggel I, Muller H. Mobile medical visual information retrieval. IEEE Trans Inf Technol Biomed 2011;16:53–61.

[50] Dialani V, Chadashvili T, Slanetz PJ. Role of imaging in neoadjuvant therapy for breast cancer. Ann Surg Oncol 2015;22:1416–24.

[51] Dolz J, Gopinath K, Yuan J, Lombaert H, Desrosiers C, Ben Ayed I. Hyperdense-net: a hyper-densely connected cnn for multi-modal image segmentation. IEEE Trans Med Imaging 2019;38:1116–26. https://doi.org/10.1109/TMI.2018.2878669.

[52] Dominguez V, Messina P, Donoso-Guzmán I, Parra D. The effect of explanations and algorithmic accuracy on visual recommender systems of artistic images. In: Proceedings of the 24th International Conference on Intelligent User Interfaces. New York, NY, USA: ACM; 2019. p. 408–16. https://doi.org/10.1145/3301275.3302274.

[53] Dove G, Halskov K, Forlizzi J, Zimmerman J. Ux design innovation: Challenges for working with machine learning as a design material. In: Proceedings of the 2017 CHI Conference on Human Factors in Computing Systems. New York, NY, USA: ACM; 2017. p. 278–88. https://doi.org/10.1145/3025453.3025739.

[54] Drnasin I, Grgić M, Gogić G. Javascript access to dicom network and objects in web browser. J Digit Imaging 2017;30:537–46.

[55] Edge R, Ford C. Clinical decision support systems for appropriate medical imaging: clinical evidence and cost-effectiveness. In: Rapid response report: summary with critical appraisal; 2019.

[56] Ehsan U, Wintersberger P, Liao QV, Mara M, Streit M, Wachter S, Riener A, Riedl MO. Operationalizing human-centered perspectives in explainable ai. In: Extended Abstracts of the 2021 CHI Conference on Human Factors in Computing Systems. New York, NY, USA: Association for Computing Machinery; 2021. p. 1–6. https://doi.org/10.1145/3411763.3441342. URL.

[57] Fails JA, Olsen Jr DR. Interactive machine learning. In: Proceedings of the 8th international conference on Intelligent user interfaces. ACM; 2003. p. 39–45.

[58] Faraji F, Gaba RC. Radiologic modalities and response assessment schemes for clinical and preclinical oncology imaging. Front Oncol 2019;9:471. https://doi.org/10.3389/fonc.2019.00471.

[59] Farrell E. Nodejs winner. CreateSpace Independent Publishing Platform; 2016.

[60] Gagnon MP, Ghandour EK, Talla PK, Simonyan D, Godin G, Labrecque M, Ouimet M, Rousseau M. Electronic health record acceptance by physicians: testing an integrated theoretical model. J Biomed Inform 2014;48:17–27. https://doi.org/10.1016/j.jbi.2013.10.010.

[61] Ghosh A. Artificial intelligence using open source bi-rads data exemplifying potential future use. J Am Coll Radiol 2019;16:64–72. https://doi.org/10.1016/j.jacr.2018.09.040.

[62] Gottapu RD, Dagli CH. Densenet for anatomical brain segmentation. Procedia Comput Sci 2018;140:179–85. https://doi.org/10.1016/j.procs.2018.10.327. URL: http://doi.acm.org/10.1145/3301275.3302290.

[63] Grier RA. How high is high? a meta-analysis of nasa-tlx global workload scores. In: Proceedings of the Human Factors and Ergonomics Society Annual Meeting. 59; 2015. p. 1727–31. https://doi.org/10.1177/1541931215591373. URL: doi:10.1177/1541931215591373.

[64] Gu D, Su K, Zhao H. A case-based ensemble learning system for explainable breast cancer recurrence prediction. Artif Intell Med 2020;107:101858. https://doi.org/

10.1016/j.artmed.2020.101858. URL: https://www.sciencedirect.com/science/article/pii/S0933365719306712.
[65] Gunning D, Stefik M, Choi J, Miller T, Stumpf S, Yang GZ. Xai – explainable artificial intelligence. <sb:contribution><sb:title>Sci</sb:title> </sb:contribution><sb:host><sb:issue><sb:series><sb:title>Robotics</sb:title></sb:series></sb:issue></sb:host> 2019;4:eaay7120. https://doi.org/10.1126/scirobotics.aay7120. URL: science.org/doi/abs/10.1126/scirobotics.aay7120.
[66] Gustin G, Macq B, Gruson D, Kieffer S. Empowerment of diabetic patients through mhealth technologies and education: development of a pilot self-management application. In: 13th International Conference on Medical Information Processing and Analysis. International Society for Optics and Photonics; 2017. 105720L.
[67] Hai J, Qiao K, Chen J, Tan H, Xu J, Zeng L, Shi D, Yan B. Fully convolutional densenet with multiscale context for automated breast tumor segmentation. J Healthc Eng 2019;2019:8415485. https://doi.org/10.1155/2019/8415485. URL: doi:10.1155/2019/8415485.
[68] Hengstler M, Enkel E, Duelli S. Applied artificial intelligence and trust—the case of autonomous vehicles and medical assistance devices. TechnolForecastSocChang 2016;105:105–20.
[69] Hoff KA, Bashir M. Trust in automation: integrating empirical evidence on factors that influence trust. Hum Factors 2015;57:407–34.
[70] Holzinger A. Interactive machine learning for health informatics: when do we need the human-in-the-loop? Brain Inform 2016;3:119–31.
[71] Holzinger A, Carrington A, Müller H. Measuring the quality of explanations: the system causability scale (scs): comparing human and machine explanations. Kunstliche intelligenz 2020;34:193–8.
[72] Holzinger A, Kieseberg P, Weippl E, Tjoa AM. Current advances, trends and challenges of machine learning and knowledge extraction: from machine learning to explainable ai. In: International Cross-Domain Conference for Machine Learning and Knowledge Extraction. Springer; 2018. p. 1–8.
[73] Holzinger A, Langs G, Denk H, Zatloukal K, Müller H. Causability and explainability of artificial intelligence in medicine. In: WIREsData Mining and Knowledge Discovery. 9; 2019. https://doi.org/10.1002/widm.1312. arXiv: https://onlinelibrary.wiley.com/doi/pdf/10.1002/widm.1312.
[74] Holzinger A, Langs G, Denk H, Zatloukal K, Müller H. Causability and explainability of artificial intelligence in medicine. Wiley Interdiscip Rev Data Min Knowl Discov 2019;9(4):e1312.
[75] Holzinger A, Plass M, Kickmeier-Rust M, Holzinger K, Crişan GC, Pintea CM, Palade V. Interactive machine learning: experimental evidence for the human in the algorithmic loop. ApplIntell 2019;49:2401–14.
[76] Huang G, Liu Z, Van Der Maaten L, Weinberger KQ. Densely connected convolutional networks. In: Proceedings of the IEEE conference on computer vision and pattern recognition; 2017. p. 4700–8. https://doi.org/10.1109/CVPR.2017.243. URL: https://ieeexplore.ieee.org/abstract/document/8099726.
[77] Hwang DK, Hsu CC, Chang KJ, Chao D, Sun CH, Jheng YC, Yarmishyn AA, Wu JC, Tsai CY, Wang ML, Peng CH, Chien KH, Kao CL, Lin TC, Woung LC, Chen SJ, Chiou SH. Artificial intelligence-based decision-making for age-related macular degeneration. Theranostics 2019;9:232–45. https://doi.org/10.7150/thno.28447. URL: https://pubmed.ncbi.nlm.nih.gov/30662564.
[78] Hwang J, Lee T, Lee H, Byun S. A clinical decision support system for sleep staging tasks with explanations from artificial intelligence: user-centered design and evaluation study. J Med Internet Res 2022;24:e28659. https://doi.org/10.2196/28659. URL: https://www.jmir.org/2022/1/e28659.
[79] Jacobs M, He J, F. Pradier M, Lam B, Ahn AC, McCoy TH, Perlis RH, Doshi-Velez F, Gajos KZ. Designing ai for trust and collaboration in time-constrained medical decisions: a sociotechnical lens. In: Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems. New York, NY, USA: Association for Computing Machinery; 2021. p. 1–14. https://doi.org/10.1145/3411764.3445385. URL: doi:10.1145/3411764.3445385.
[80] Jiménez Gaona Y, Rodriguez-Alvarez MJ, Espino-Morato H, Castillo Malla D, Lakshminarayanan V. Densenet for breast tumor classification in mammographic images. In: Rojas I, Castillo-Secilla D, Herrera LJ, Pomares H, editors. Bioengineering and biomedical signal and image processing. Cham: Springer International Publishing; 2021. p. 166–76. https://doi.org/10.1007/978-3-030-88163-4_16.
[81] Jodogne S. The orthanc ecosystem for medical imaging. J Digit Imaging 2018;31:341–52. https://doi.org/10.1007/s10278-018-0082-y.
[82] Johnson WL, Lester JC. Face-to-face interaction with pedagogical agents, twenty years later. IntJArtifIntellEduc 2016;26:25–36. https://doi.org/10.1007/s40593-015-0065-9. URL: 10.1007/s40593-015-0065-9.
[83] Keay T, Conway CM, O'Flaherty N, Hewitt SM, Shea K, Gavrielides MA. Reproducibility in the automated quantitative assessment of her2/neu for breast cancer. J Pathol Inform 2013;4. https://doi.org/10.4103/2153-3539.115879.
[84] Khairat S, Marc D, Crosby W, Al Sanousi A. Reasons for physicians not adopting clinical decision support systems: critical analysis. JMIR Med Inform 2018;6:e24.
[85] Kingma DP, Ba J. Adam: a method for stochastic optimization. 2014. arXiv preprint arXiv:1412.6980.
[86] Kocielnik R, Amershi S, Bennett PN. Will you accept an imperfect ai?: Exploring designs for adjusting end-user expectations of ai systems. In: Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems. 14. New York, NY, USA: ACM; 2019. p. 1–411. https://doi.org/10.1145/3290605.3300641.
[87] Kohli A, Jha S. Why cad failed in mammography. J Am Coll Radiol 2018;15:535–7.
[88] Lau JJ, Gayen S, Ben Abacha A, Demner-Fushman D. A dataset of clinically generated visual questions and answers about radiology images. SciData 2018;5:180251. https://doi.org/10.1038/sdata.2018.251. URL: 10.1038/sdata.2018.251.
[89] Lee MH, Siewiorek DP, Smailagic A, Bernardino A, Bermúdez i Badia S. A human-ai collaborative approach for clinical decision making on rehabilitation assessment. In: Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems. New York, NY, USA: Association for Computing Machinery; 2021. p. 1–14. https://doi.org/10.1145/3411764.3445472.
[90] Lencastre H. Breast cancer multimoladity scalable interactions. URL: http://rgdoi.net/10.13140/RG.2.2.35800.24329/4. Avenida Rovisco Pais 1, 1049-001 Lisboa - Portugal (EU): Instituto Superior Técnico; 2020. https://doi.org/10.13140/RG.2.2.35800.24329. improving a framework with novel interactive techniques which enhances the diagnostic process.
[91] Leung L, Chen C. E-health/m-health adoption and lifestyle improvements: exploring the roles of technology readiness, the expectation-confirmation model, and health-related information activities. Telecommun Policy 2019;43:563–75. https://doi.org/10.1016/j.telpol.2019.01.005. URL: https://www.sciencedirect.com/science/article/pii/S0308596117304263.
[92] Levy A, Agrawal M, Satyanarayan A, Sontag D. Assessing the impact of automated suggestions on decision making: domain experts mediate model errors but take less initiative. In: Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems. New York, NY, USA: Association for Computing Machinery; 2021. p. 1–13. https://doi.org/10.1145/3411764.3445522. URL: doi:10.1145/3411764.3445522.
[93] Li W, Zhou X, Yang Q. Designing medical artificial intelligence for in- and out-groups. Comput Hum Behav 2021;124:106929. https://doi.org/10.1016/j.chb.2021.106929. URL: https://www.sciencedirect.com/science/article/pii/S0747563221002521.
[94] Liikkanen LA. In: The data-driven design era in professional web design. interactions. 24; 2017. p. 52–7.
[95] Lu Z, Yin M. Human reliance on machine learning models when performance feedback is limited: heuristics and risks. In: Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems. New York, NY, USA: Association for Computing Machinery; 2021. p. 1–16. https://doi.org/10.1145/3411764.3445562. URL: 10.1145/3411764.3445562.
[96] Maicas G, Bradley AP, Nascimento JC, Reid I, Carneiro G. Training medical image analysis systems like radiologists. In: Frangi AF, Schnabel JA, Davatzikos C, Alberola-López C, Fichtinger G, editors. Medical image computing and computer assisted intervention – MICCAI 2018. Cham: Springer International Publishing; 2018. p. 546–54.
[97] Maicas G, Leonardi M, Avery J, Panuccio C, Carneiro G, Hull ML, Condous G. Deep learning to diagnose pouch of Douglas obliteration with ultrasound sliding sign. Reprod Fertil 2021;2:236–43. https://doi.org/10.1530/RAF-21-0031. URL: doi:10.1145/3411764.3445385.
[98] Massat MB. A promising future for ai in breast cancer screening. Appl Radiol 2018;47:22–5.
[99] Mathews A, Marc D. Usability evaluation of laboratory information systems. JPatholInform 2017;8.
[100] McKinney SM, Sieniek M, Godbole V, Godwin J, Antropova N, Ashrafian H, Back T, Chesus M, Corrado GS, Darzi A, Etemadi M, Garcia-Vicente F, Gilbert FJ, Halling-Brown M, Hassabis D, Jansen S, Karthikesalingam A, Kelly CJ, King D, Ledsam JR, Melnick D, Mostofi H, Peng L, Reicher JJ, Romera-Paredes B, Sidebottom R, Suleyman M, Tse D, Young KC, De Fauw J, Shetty S. International evaluation of an ai system for breast cancer screening. Nature 2020;577:89–94. https://doi.org/10.1038/s41586-019-1799-6.
[101] Medley D, Santiago C, Nascimento JC. Cycoseg: a cyclic collaborative framework for automated medical image segmentation. IEEE Trans Pattern Anal Mach Intell 2021;1–1. https://doi.org/10.1109/TPAMI.2021.3113077.
[102] Miller A. The intrinsically linked future for human and artificial intelligence interaction. JBig Data 2019;6:38.
[103] Miller T. Explanation in artificial intelligence: insights from the social sciences. ArtifIntell 2018;267:1–38. https://doi.org/10.1016/j.artint.2018.07.007.
[104] Mou Y, Xu K. The media inequality: comparing the initial human-human and human-ai social interactions. ComputHumBehav 2017;72:432–40. https://doi.org/10.1016/j.chb.2017.02.067.
[105] Mourão N, Calisto FM, Nascimento JC. Mimbcd-ui: Ai visual explanation architecture. 2020. https://doi.org/10.13140/RG.2.2.11568.30723. URL: http://rgdoi.net/10.13140/RG.2.2.11568.30723.
[106] Murray L, Gopinath D, Agrawal M, Horng S, Sontag D, Karger DR. Medknowts: unified documentation and information retrieval for electronic health records. In: The 34th Annual ACM Symposium on User Interface Software and Technology. New York, NY, USA: Association for Computing Machinery; 2021. p. 1169–83. https://doi.org/10.1145/3472749.3474814.
[107] Myers BA. A brief history of human-computer interaction technology. Interactions 1998;5:44–54. https://doi.org/10.1145/274430.274436. URL: 10.1145/274430.274436.
[108] Mourão Nádia. Master project: 2D breast cancer diagnosis explainable visualizations. URL: http://rgdoi.net/10.13140/RG.2.2.31605.93928/4. Avenida Rovisco Pais 1, 1049-001 Lisboa - Portugal (EU): Instituto Superior Técnico; 2020. https://doi.org/10.13140/RG.2.2.31605.93928/4. introduction of explainable methods on medical decision making.
[109] Mourão Nádia. 2D breast cancer diagnosis explainable visualizations. URL: http://rgdoi.net/10.13140/RG.2.2.18809.80480. Avenida Rovisco Pais 1, 1049-001 Lisboa - Portugal (EU): Instituto Superior Técnico; 2021. https://doi.org/10.13140/RG.2.2.18809.80480. introduction of explainable methods on medical decision-making.
[110] K. Nandy Y. Liu D. Mott K. Meaburn T. Misteli S.J. Lockett P.R. Gudla ,. Mipipeline (microscopy pipeline): A user friendly software environment for microscopy image analysis and informatics.

[111] Nassif AB, Talib MA, Nasir Q, Afadar Y, Elgendy O. Breast cancer detection using artificial intelligence techniques: a systematic literature review. Artif Intell Med 2022;127:102276. https://doi.org/10.1016/j.artmed.2022.102276. URL: https://www.sciencedirect.com/science/article/pii/S0933365722000410.
[112] Osman Andersen T, Nunes F, Wilcox L, Kaziunas E, Matthiesen S, Magrabi F. chapter 8. In: Realizing AI in healthcare: challenges appearing in the wild. New York, NY, USA: Association for Computing Machinery; 2021. p. 1687–92. https://doi.org/10.1145/3411763.3441347. URL.
[113] Paszke A, Gross S, Massa F, Lerer A, Bradbury J, Chanan G, et al. Pytorch: an imperative style, high-performance deep learning library. In: Wallach H, Larochelle H, Beygelzimer A, d'Alché-Buc F, Fox E, Garnett R, editors. Advances in neural information processing systems. 32. Curran Associates, Inc.; 2019. p. 8024–35. URL: https://arxiv.org/abs/1912.01703.
[114] Price WN. Big data and black-box medical algorithms. Sci Transl Med 2018;10: eaao5333. https://doi.org/10.1126/scitranslmed.aao5333. URL: https://www.science.org/doi/abs/10.1126/scitranslmed.aao5333.
[115] Quinn TP, Jacobs S, Senadeera M, Le V, Coghlan S. The three ghosts of medical ai: can the black-box present deliver? Artif Intell Med 2022;124:102158. https://doi.org/10.1016/j.artmed.2021.102158.
[116] Raja K, Patrick M, Elder JT, Tsoi LC. Machine learning workflow to enhance predictions of adverse drug reactions (adrs) through drug-gene interactions: application to drugs for cutaneous diseases. Sci Rep 2017;7:3690. https://doi.org/10.1038/s41598-017-03914-3. URL: https://www.nature.com/articles/s41598-017-03914-3.
[117] Ramkumar A, Stappers PJ, Niessen WJ, Adebahr S, Schimek-Jasch T, Nestle U, Song Y. Using GOMS and NASA-TLX to evaluate human-computer interaction process in interactive segmentation. Int JHumComputInteract 2017;33:123–34. https://doi.org/10.1080/10447318.2016.1220729. URL: https://www.tandfonline.com/doi/full/10.1080/10447318.2016.1220729.
[118] Redondo A, Comas M, Macià F, Ferrer F, Murta-Nascimento C, Maristany MT, Molins E, Sala M, Castells X. Inter- and intraradiologist variability in the bi-rads assessment and breast density categories for screening mammograms. Br J Radiol 2012;85:1465–70. https://doi.org/10.1259/bjr/21256379. doi:10.1259/bjr/21256379. 22993385[pmid].
[119] Riasatian A, Babaie M, Maleki D, Kalra S, Valipour M, Hemati S, Zaveri M, Safarpoor A, Shafiei S, Afshari M, Rasoolijaberi M, Sikaroudi M, Adnan M, Shah S, Choi C, Damaskinos S, Campbell CJ, Diamandis P, Pantanowitz L, Kashani H, Ghodsi A, Tizhoosh H. Fine-tuning and training of densenet for histopathology image representation using tcga diagnostic slides. Med Image Anal 2021;70: 102032. https://doi.org/10.1016/j.media.2021.102032.
[120] Rowell MR, Corl FM, Johnson PT, Fishman EK. Internet-based dissemination of educational audiocasts: a primer in podcasting-how to do it. Am J Roentgenol 2006;186:1792–6.
[121] Rybiałek A, Jeleń Ł. Application of densenets for classification of breast cancer mammograms. In: Saeed K, Dvorský J, editors. Computer information systems and industrial management. Cham: Springer International Publishing; 2020. p. 266–77.
[122] Samek W, Montavon G, Vedaldi A, Hansen LK, Muller KR. Explainable AI: interpreting, explaining and visualizing deep learning. 1st ed. Springer Publishing Company,Incorporated; 2019.
[123] Santiago C, Medley DO, Marques JS, Nascimento JC. Model-agnostic temporal regularizer for object localization using motion fields. IEEE Trans Image Process 2022;31:2478–87. https://doi.org/10.1109/TIP.2022.3155947.
[124] Savage N. Digital assistants aid disease diagnosis. Nature 2019;573:S98–9. https://doi.org/10.1038/d41586-019-02870-4. URL: 10.1038/d41586-019-02870-4.
[125] Schaekermann M, Beaton G, Sanoubari E, Lim A, Larson K, Law E. Ambiguity-aware ai assistants for medical data analysis. In: Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems. New York, NY, USA: Association for Computing Machinery; 2020. p. 1–14. https://doi.org/10.1145/3313831.3376506.
[126] Schaekermann M, Cai CJ, Huang AE, Sayres R. Expert discussions improve comprehension of difficult cases in medical image assessment. In: Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems. New York, NY, USA: Association for Computing Machinery; 2020. p. 1–13. https://doi.org/10.1145/3313831.3376290. URL: doi:10.1145/3313831.3376290.
[127] Seifabadi R, Cheng A, Malik B, Kishimoto S, Wiskin J, Munasinghe J, Negussie AH, Bakhutashvili I, Krishna MC, Choyke P, et al. Correlation of ultrasound tomography to mri and pathology for the detection of prostate cancer. In: Medical Imaging 2019: Ultrasonic Imaging and Tomography. International Society for Optics and Photonics; 2019. 109550C.
[128] Seref B, Bostanci E. Performance of naïve and complement naïve Bayes algorithms based on accuracy, precision and recall performance evaluation criterions. IntJComput 2019;8:75–92.
[129] Shah P, Kendall F, Khozin S, Goosen R, Hu J, Laramie J, Ringel M, Schork N. Artificial intelligence and machine learning in clinical development: a translational perspective. NPJ DigMed 2019;2:69.
[130] Sidenko I, Filina K, Kondratenko G, Chabanovskyi D, Kondratenko Y. Eye-tracking technology for the analysis of dynamic data. In: 2018 IEEE 9th International Conference on Dependable Systems, Services and Technologies (DESSERT). IEEE; 2018. p. 479–84.
[131] Suh YJ, Jung J, Cho BJ. Automated breast cancer detection in digital mammograms of various densities via deep learning. JPersonalized Med 2020;10. https://doi.org/10.3390/jpm10040211.
[132] Topol EJ. High-performance medicine: the convergence of human and artificial intelligence. Nat Med 2019;25:44.
[133] Tran TNT, Felfernig A, Trattner C, Holzinger A. Recommender systems in the healthcare domain: state-of-the-art and research issues. J Intell Inform Syst 2020. https://doi.org/10.1007/s10844-020-00633-6.
[134] Urban T, Ziegler E, Lewis R, Hafey C, Sadow C, Van den Abbeele AD, Harris GJ. LesionTracker: extensible open-source zero-footprint web viewer for cancer imaging research and clinical trials. Cancer Res 2017;77:119–22.
[135] Wang D, Wang L, Zhang Z, Wang D, Zhu H, Gao Y, Fan X, Tian F. "brilliant ai doctor" in rural clinics: challenges in ai-powered clinical decision support system deployment. In: Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems. New York, NY, USA: Association for Computing Machinery; 2021. p. 1–18. https://doi.org/10.1145/3411764.3445432. URL: 10.1145/3411764.3445432.
[136] Wang H, Li Y, Khan SA, Luo Y. Prediction of breast cancer distant recurrence using natural language processing and knowledge-guided convolutional neural network. Artif Intell Med 2020;110:101977. https://doi.org/10.1016/j.artmed.2020.101977. URL: https://www.sciencedirect.com/science/article/pii/S0933365720312422.
[137] Weisz JD, Jain M, Joshi NN, Johnson J, Lange I. Bigbluebot: teaching strategies for successful human-agent interactions. In: Proceedings of the 24th International Conference on Intelligent User Interfaces. New York, NY, USA: ACM; 2019. p. 448–59. https://doi.org/10.1145/3301275.3302290. URL: http://doi.acm.org/10.1145/3301275.3302290.
[138] Wobbrock JO, Findlater L, Gergle D, Higgins JJ. The aligned rank transform for nonparametric factorial analyses using only anova procedures. In: Proceedings of the SIGCHI Conference on Human Factors in Computing Systems. New York, NY, USA: ACM; 2011. p. 143–6. https://doi.org/10.1145/1978942.1978963.
[139] Xie Y, Chen M, Kao D, Gao G, Chen XA. Chexplain: enabling physicians to explore and understand data-driven, ai-enabled medical imaging analysis. In: Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems. New York, NY, USA: Association for Computing Machinery; 2020. p. 1–13. https://doi.org/10.1145/3313831.3376807.
[140] Yang Q, Steinfeld A, Zimmerman J. Unremarkable ai: fitting intelligent decision support into critical, clinical decision-making processes. In: Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems. New York, NY, USA: Association for Computing Machinery; 2019. p. 1–11. https://doi.org/10.1145/3290605.3300468. URL: 10.1145/3290605.3300468.
[141] Yang Q, Zimmerman J, Steinfeld A, Carey L, Antaki JF. Investigating the heart pump implant decision process: opportunities for decision support tools to help. In: Proceedings of the 2016 CHI Conference on Human Factors in Computing Systems. ACM; 2016. p. 4477–88.
[142] Zhang M, Pavlicek W, Panda A, Langer SG, Morin R, Fetterly KA, Paden R, Hanson J, Wu LW, Wu T. Dicom index tracker enterprise: advanced system for enterprise-wide quality assurance and patient safety monitoring. In: Medical Imaging 2015: PACS and Imaging Informatics: Next Generation and Innovations. International Society for Optics and Photonics; 2015. 94180L.
[143] Zhong Z, Zheng M, Mai H, Zhao J, Liu X. Cancer image classification based on DenseNet model. JPhysConfSer 2020;1651:012143. https://doi.org/10.1088/1742-6596/1651/1/012143.
[144] Zimmerman J. Commentary: critique of contesting control: how an excellent exemplar of artist-led hci research surfaces unresolved issues. HumComputInteract 2021;36:390–3. https://doi.org/10.1080/07370024.2020.1816832.
[145] Zimmerman J, Oh C, Yildirim N, Kass A, Tung T, Forlizzi J. Ux designers pushing ai in the enterprise: a case for adaptive uis. Interactions 2020;28:72–7. https://doi.org/10.1145/3436954. URL: 10.1145/3436954.